



US012405504B2

(12) **United States Patent**
Jack et al.

(10) **Patent No.:** **US 12,405,504 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ANGLED BUS BAR**

(56) **References Cited**

(71) Applicant: **View, Inc.**, Milpitas, CA (US)

U.S. PATENT DOCUMENTS

(72) Inventors: **Gordon Edmund Jack**, San Jose, CA (US); **Anshu Ajit Pradhan**, Collierville, TN (US); **Kaustubh Yatindra Nadkarni**, Memphis, TN (US)

4,878,743 A 11/1989 Aikawa et al.
5,076,673 A 12/1991 Lynam et al.
(Continued)

FOREIGN PATENT DOCUMENTS

(73) Assignee: **View Operating Corporation**, San Jose, CA (US)

CN 1936711 A 3/2007
CN 102237446 A 11/2011
(Continued)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

OTHER PUBLICATIONS

IN First Examination Report dated Sep. 27, 2021, in application No. IN202138008053.

(Continued)

(21) Appl. No.: **17/933,115**

(22) Filed: **Sep. 18, 2022**

(65) **Prior Publication Data**

US 2023/0077475 A1 Mar. 16, 2023

Primary Examiner — Jie Lei

(74) *Attorney, Agent, or Firm* — Cha & Reiter, LLC

Related U.S. Application Data

(63) Continuation of application No. 17/930,883, filed on Sep. 9, 2022, which is a continuation of application No. 16/032,842, filed on Jul. 11, 2018, now Pat. No. 11,480,841, which is a continuation of application No. 13/452,032, filed on Apr. 20, 2012, now Pat. No. 10,429,712.

(51) **Int. Cl.**
G02F 1/155 (2006.01)

(52) **U.S. Cl.**
CPC **G02F 1/155** (2013.01)

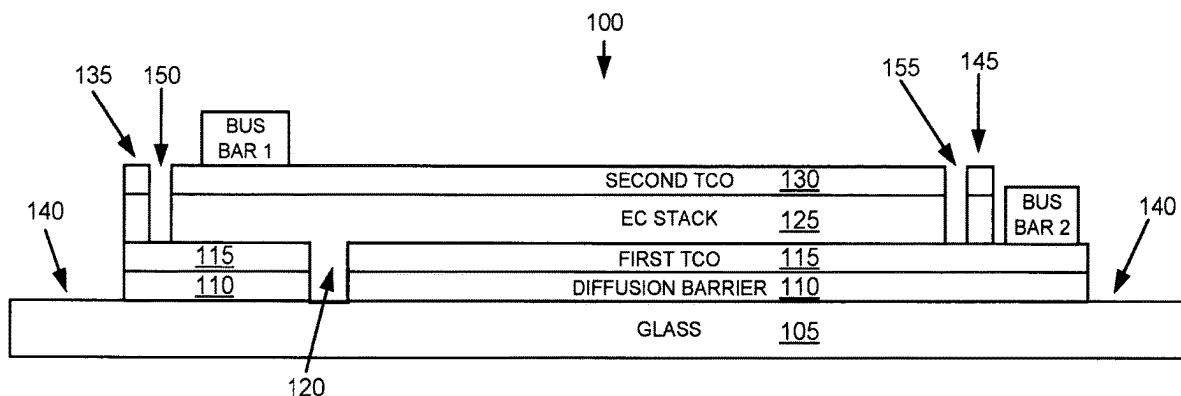
(58) **Field of Classification Search**
CPC G02F 1/155; G02F 1/1525; G02F 1/1523; G02F 1/163; G02F 1/153; G02B 5/32
USPC 359/265, 266, 270, 275, 900; 427/58; 174/250

See application file for complete search history.

(57) **ABSTRACT**

This disclosure provides configurations, methods of use, and methods of fabrication for a bus bar of an optically switchable device. In one aspect, an apparatus includes a substrate and an optically switchable device disposed on a surface of the substrate. The optically switchable device has a perimeter with at least one corner including a first side, a second side, and a first vertex joining the first side and the second side. A first bus bar and a second bus bar are affixed to the optically switchable device and configured to deliver current and/or voltage for driving switching of the device. The first bus bar is proximate to the corner and includes a first arm and a second arm having a configuration that substantially follows the shape of the first side, the first vertex, and the second side of the corner.

17 Claims, 25 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

5,140,455	A	8/1992	Varaprasad et al.	7,869,114	B2	1/2011	Valentin et al.
5,142,407	A	8/1992	Varaprasad et al.	7,894,119	B2	2/2011	Valentin et al.
5,145,609	A	9/1992	Varaprasad et al.	7,929,194	B2	4/2011	Legois et al.
5,151,816	A	9/1992	Varaprasad et al.	7,952,785	B2	5/2011	Karmhag et al.
5,187,607	A	2/1993	Endo et al.	8,035,882	B2	10/2011	Fanton et al.
5,233,461	A	8/1993	Dorman et al.	10,429,712	B2	10/2019	Jack et al.
5,239,405	A	8/1993	Varaprasad et al.	10,481,458	B2	11/2019	Dixit et al.
5,340,503	A	8/1994	Varaprasad et al.	11,333,948	B2	5/2022	Dixit et al.
5,379,146	A	1/1995	Defendini	11,480,841	B2	10/2022	Jack et al.
5,471,338	A	11/1995	Yu et al.	2002/0021481	A1	2/2002	Lin et al.
5,472,643	A	12/1995	Varaprasad et al.	2002/0041443	A1	4/2002	Varaprasad et al.
5,500,760	A	3/1996	Varaprasad et al.	2002/0075552	A1	6/2002	Poll et al.
5,567,360	A	10/1996	Varaprasad et al.	2002/0135881	A1	9/2002	Rukavina et al.
5,657,149	A	8/1997	Buffat et al.	2002/0149829	A1	10/2002	Mochizuka et al.
5,657,150	A	8/1997	Kallman et al.	2003/0227663	A1	12/2003	Agrawal et al.
5,668,663	A	9/1997	Varaprasad et al.	2004/0047050	A1	3/2004	Bauer et al.
5,724,175	A	3/1998	Hichwa et al.	2004/0257633	A1	12/2004	Agrawal et al.
5,724,187	A	3/1998	Varaprasad et al.	2005/0002081	A1	1/2005	Beteille et al.
5,805,330	A	9/1998	Byker et al.	2005/0056257	A1	3/2005	Yoshioka et al.
5,805,367	A	9/1998	Kanazawa	2006/0077511	A1	4/2006	Poll et al.
5,953,150	A	9/1999	Smarto et al.	2006/0187608	A1	8/2006	Stark
5,969,847	A	10/1999	Coleman et al.	2007/0002422	A1	1/2007	O'Shaughnessy
5,985,184	A	11/1999	Lynam	2007/0020442	A1	1/2007	Giron et al.
5,995,271	A	11/1999	Zieba et al.	2007/0067048	A1	3/2007	Bechtel et al.
6,001,487	A	12/1999	Ladang et al.	2007/0103761	A1	5/2007	Giron et al.
6,002,511	A	12/1999	Varaprasad et al.	2007/0133078	A1	6/2007	Fanton et al.
6,031,201	A	2/2000	Amako et al.	2008/0074724	A1	3/2008	Agrawal et al.
6,045,643	A	4/2000	Byker et al.	2008/0115428	A1	5/2008	Schlam et al.
6,045,896	A	4/2000	Boire et al.	2009/0058295	A1	3/2009	Auday et al.
6,055,088	A	4/2000	Fix et al.	2009/0067031	A1	3/2009	Piroux et al.
6,055,089	A	4/2000	Schulz et al.	2009/0097098	A1	4/2009	Piroux
6,118,573	A	9/2000	Kubo et al.	2009/0110918	A1	4/2009	Jacquiod et al.
6,143,209	A	11/2000	Lynam	2009/0114928	A1	5/2009	Messere et al.
6,154,306	A	11/2000	Varaprasad et al.	2009/0130409	A1	5/2009	Reutler et al.
6,166,849	A	12/2000	Coleman et al.	2009/0148642	A1	6/2009	Mausser et al.
6,204,953	B1	3/2001	Zieba et al.	2009/0174300	A1	7/2009	Jousse et al.
6,207,083	B1	3/2001	Varaprasad et al.	2009/0181203	A1	7/2009	Valentin et al.
6,244,716	B1	6/2001	Steenwyk et al.	2009/0251758	A1	10/2009	Valentin et al.
6,261,641	B1	7/2001	Zieba et al.	2009/0262411	A1	10/2009	Karmhag et al.
6,317,248	B1 *	11/2001	Agrawal G02F 1/155 348/148	2009/0297806	A1	12/2009	Dawson-Elli et al.
6,337,758	B1	1/2002	Beteille et al.	2009/0323160	A1	12/2009	Egerton et al.
6,433,913	B1	8/2002	Bauer et al.	2010/0067090	A1	3/2010	Egerton et al.
6,471,360	B2	10/2002	Rukavina et al.	2010/0208326	A1	8/2010	Kwak et al.
6,515,787	B1	2/2003	Westfall et al.	2010/0243427	A1	9/2010	Kozlowski et al.
6,529,308	B2	3/2003	Beteille et al.	2011/0043885	A1	2/2011	Lamine et al.
6,535,126	B2	3/2003	Lin et al.	2011/0048614	A1	3/2011	Veerasamy
6,559,411	B2	5/2003	Borgeson et al.	2011/0051221	A1	3/2011	Veerasamy
6,561,460	B2	5/2003	Rukavina et al.	2011/0059275	A1	3/2011	Stark
6,783,099	B2	8/2004	Rukavina et al.	2011/0211246	A1	9/2011	Agrawal et al.
6,795,226	B2	9/2004	Agrawal et al.	2011/0216389	A1	9/2011	Piroux et al.
6,798,556	B2	9/2004	Tench et al.	2011/0260961	A1	10/2011	Burdis
6,822,778	B2	11/2004	Westfall et al.	2011/0267672	A1	11/2011	Sbar et al.
6,853,472	B2	2/2005	Warner et al.	2011/0299149	A1	12/2011	Park et al.
6,862,125	B2	3/2005	Warner et al.	2012/0019889	A1	1/2012	Lamine et al.
6,919,530	B2	7/2005	Borgeson et al.	2012/0026573	A1	2/2012	Collins et al.
6,995,891	B2	2/2006	Agrawal et al.	2012/0062975	A1	3/2012	Mehtani et al.
7,002,720	B2	2/2006	Beteille et al.	2012/0062976	A1	3/2012	Burdis
7,004,592	B2	2/2006	Varaprasad et al.	2012/0194895	A1	8/2012	Podbelski et al.
7,033,655	B2	4/2006	Beteille et al.	2012/0200908	A1	8/2012	Bergh et al.
7,130,101	B2	10/2006	Rukavina et al.	2012/0300280	A1	11/2012	Murphy et al.
7,133,181	B2	11/2006	Greer	2012/0327499	A1	12/2012	Parker et al.
7,173,750	B2	2/2007	Rukavina	2013/0016417	A1	1/2013	Veerasamy
7,202,987	B2	4/2007	Varaprasad et al.	2013/0032084	A1	2/2013	Sivaramakrishnan et al.
7,230,748	B2	6/2007	Giron et al.	2013/0037992	A1	2/2013	Milshtein et al.
7,248,392	B2	7/2007	Rukavina et al.	2013/0278988	A1	10/2013	Jack et al.
7,277,215	B2	10/2007	Greer	2014/0007418	A1	1/2014	Song et al.
7,300,166	B2	11/2007	Agrawal et al.	2016/0097960	A1	4/2016	Dixit et al.
7,317,106	B2	1/2008	Warner et al.	2019/0011793	A1	1/2019	Jack et al.
7,324,261	B2	1/2008	Tonar et al.	2020/0041860	A1	2/2020	Dixit et al.
7,362,491	B2	4/2008	Busick et al.	2022/0244609	A1	8/2022	Dixit et al.
7,477,439	B2	1/2009	Tonar et al.	2023/0004057	A1	1/2023	Jack et al.
7,531,101	B2	5/2009	Beteille	2023/0086158	A1	3/2023	Jack et al.
7,710,671	B1	5/2010	Kwak et al.				
7,719,751	B2	5/2010	Egerton et al.				

FOREIGN PATENT DOCUMENTS

CN	102460291	A	5/2012
CN	102460292	A	5/2012
CN	102460630	A	5/2012
CN	103238107	A	8/2013

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	103370649	A	10/2013
CN	103492940	A	1/2014
DE	102006042538	A1	3/2008
EP	0356099	A2	2/1990
EP	0470867	A2	2/1992
EP	0851271	A2	7/1998
EP	0950568	A1	10/1999
EP	1420287	A1	5/2004
EP	2348357	A2	7/2011
GB	2190760	A	11/1987
JP	2004537846	A	12/2004
KR	20120038439	A	4/2012
TW	201235757	A	9/2012
WO	WO-2005076061	A1	8/2005
WO	WO-2007100921	A2	9/2007
WO	WO-2008043951	A2	4/2008
WO	WO-2009145876	A1	12/2009
WO	WO-2009148861	A2	12/2009
WO	WO-2011010067	A2	1/2011
WO	WO-2011028253	A2	3/2011
WO	WO-2011028254	A2	3/2011
WO	WO-2011050291	A2	4/2011
WO	WO-2011109688	A1	9/2011
WO	WO-2012145155	A1	10/2012
WO	WO-2014205014	A1	12/2014

OTHER PUBLICATIONS

Notice of Allowance with Search Report, dated Dec. 5, 2019 in CN Application No. 201480035059.2, with English translation.

U.S. Final Office Action dated Dec. 15, 2023 in U.S. Appl. No. 17/993,577.

U.S. Non-Final Office Action dated Apr. 18, 2023 in U.S. Appl. No. 17/993,577.

U.S. Non-Final Office Action dated Feb. 21, 2024 in U.S. Appl. No. 17/930,883.

U.S. Restriction Requirement dated Feb. 9, 2023 in U.S. Appl. No. 17/993,577.

U.S. Restriction requirement dated Jan. 4, 2024, in U.S. Appl. No. 17/658,432.

U.S. Restriction requirement dated Oct. 23, 2023 in U.S. Appl. No. 17/930,883.

CA Office Action dated Jan. 21, 2022, in Application No. CA2915525.
Canadian Office Action dated Jul. 3, 2020, in Canadian Patent Application No. 2,915,525.

Canadian Office Action dated Mar. 31, 2021, in Canadian Patent Application No. 2,915,525.

Chinese Office Action dated Apr. 15, 2019 in CN Application No. 201480035059.2.

Chinese Office Action dated Jul. 20, 2017 for CN Application No. 201380020654.4.

Chinese Office Action dated Jun. 4, 2018 in CN Application No. 201480035059.2.

Chinese Office Action dated Mar. 20, 2018 in CN Application No. 201380020654.4.

Chinese Office Action dated Mar. 5, 2019 in CN Application No. 201380020654.4.

Chinese Office Action dated Sep. 28, 2016 in CN Application No. 201380020654.4.

CN Office Action dated May 16, 2022, in Application No. CN201910816267.0 with English translation.

CN Office Action dated Sep. 1, 2021, in CN Application No. CN201910816267.0 with English translation.

European Examination Report dated Nov. 16, 2016 for EP Application No. 13778088.8.

European Extended Search Report dated Jul. 24, 2018 for EP Application No. 17205976.8.

European Extended Search Report dated Nov. 2, 2016 for EP Application No. 14814624.4.

European Office Action dated Apr. 22, 2021 for EP Application No. 14814624.4.

European Office Action dated Aug. 31, 2017 for EP Application No. 14814624.4.

European Office Action dated Jun. 28, 2019 for EP Application No. 14814624.4.

European Office Action dated Jun. 3, 2020 for EP Application No. 14814624.4.

European Office Action dated May 4, 2018 for EP Application No. 14814624.4.

European Office Action dated Nov. 12, 2018 for EP Application No. 14814624.4.

European Supplementary Search Report dated Jun. 25, 2015 for EP09815048.5.

European Supplementary Search Report dated Oct. 27, 2015 for EP Application No. 13778088.8.

IN Office Action dated Sep. 9, 2022, in Application No. IN202238020340.

Indian Office Action dated Mar. 11, 2020, for Indian Patent Application No. 4020/KOLNP/2015.

International Preliminary Report on Patentability dated Dec. 30, 2015 from PCT/US2014/042819.

International Preliminary Report on Patentability dated Mar. 31, 2011 from PCT/US2009/56928.

International Preliminary Report on Patentability dated Oct. 30, 2014 from PCTUS2013/037115.

International Search Report and Written Opinion dated Jul. 22, 2013 from PCTUS2013/037115.

International Search Report and Written Opinion dated May 4, 2010 from PCT/US2009/56928.

International Search Report and Written Opinion dated Oct. 21, 2014 from PCT/US2014/042819.

Notice of Allowance with Search Report, dated Dec. 5, 2019 in CN Application No. 201480035059.2, no translation.

Russian Office Action dated Jul. 30, 2018 for RU Application No. 2016101089.

Russian Office Action dated Mar. 6, 2018 for RU Application No. 2016101089.

Taiwanese Office Action dated Aug. 3, 2018 for TW Application No. 103120947.

Taiwanese Office Action dated Dec. 27, 2017 for TW Application No. 103120947.

Taiwanese Office Action dated Jan. 23, 2019 for TW Application No. 103120947.

U.S. Corrected Notice of Allowance dated Sep. 28, 2022 in U.S. Appl. No. 16/032,842.

U.S. Final Office Action dated Apr. 20, 2015 for U.S. Appl. No. 13/452,032.

U.S. Final Office Action dated Aug. 29, 2017 for U.S. Appl. No. 14/893,505.

U.S. Final Office Action dated Dec. 13, 2021 for U.S. Appl. No. 16/032,842.

U.S. Final Office Action dated Jan. 7, 2019 for U.S. Appl. No. 14/893,505.

U.S. Final Office Action dated May 2, 2014 for U.S. Appl. No. 13/452,032.

U.S. Final Office Action dated Sep. 1, 2016 for U.S. Appl. No. 13/452,032.

U.S. Notice of Allowance dated Apr. 13, 2022 in U.S. Appl. No. 16/579,778.

U.S. Notice of Allowance dated Apr. 20, 2022 in U.S. Appl. No. 16/579,778.

U.S. Notice of Allowance dated Jan. 10, 2022, in U.S. Appl. No. 16/579,778.

U.S. Notice of Allowance dated Jan. 28, 2019 for U.S. Appl. No. 13/452,032.

U.S. Notice of Allowance dated Jun. 13, 2022, in U.S. Appl. No. 16/032,842.

U.S. Notice of Allowance dated Jun. 26, 2019 for U.S. Appl. No. 14/893,505.

U.S. Notice of Allowance dated Mar. 11, 2010 for U.S. Appl. No. 12/212,482.

(56)

References Cited

OTHER PUBLICATIONS

U.S. Notice of Allowance dated May 24, 2019 for U.S. Appl. No. 13/452,032.

U.S. Office Action dated Apr. 17, 2020 for U.S. Appl. No. 16/032,842.

U.S. Office Action dated Dec. 8, 2015 for U.S. Appl. No. 13/452,032.

U.S. Office Action dated Jun. 23, 2017 for U.S. Appl. No. 13/452,032.

U.S. Office Action dated Jun. 4, 2021 for U.S. Appl. No. 16/579,778.

U.S. Office Action dated May 13, 2021 for U.S. Appl. No. 16/032,842.

U.S. Office Action dated May 17, 2018 for U.S. Appl. No. 14/893,505.

U.S. Office Action dated Nov. 13, 2014 for U.S. Appl. No. 13/452,032.

U.S. Office Action dated Nov. 21, 2013 for U.S. Appl. No. 13/452,032.

U.S. Office Action dated Nov. 3, 2016 for U.S. Appl. No. 14/893,505.

U.S. Office Action dated Oct. 5, 2020 for U.S. Appl. No. 16/579,778.

U.S. Office Action dated Sep. 25, 2020 for U.S. Appl. No. 16/032,842.

U.S. Appl. No. 17/930,883, inventor Jack et al., filed on Sep. 9, 2022.

CA Office Action dated Nov. 17, 2022, in Application No. CA2915525.

U.S. Final Office Action dated Oct. 23, 2024 in U.S. Appl. No. 17/930,883.

U.S. Non-Final Office Action dated Aug. 1, 2024 in U.S. Appl. No. 17/993,577.

* cited by examiner

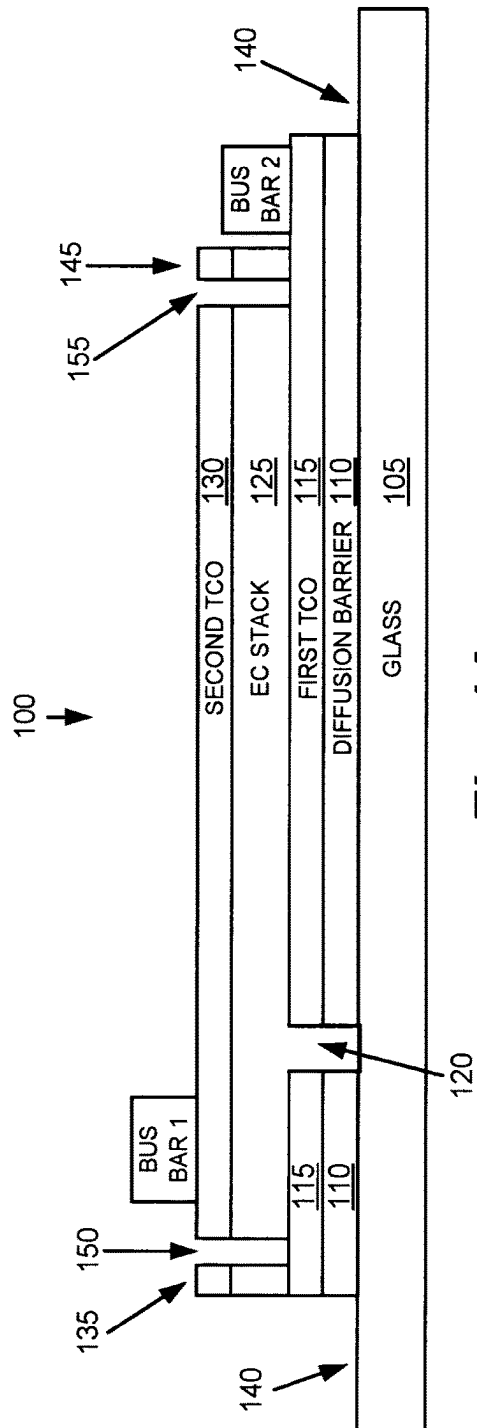


Fig. 1A

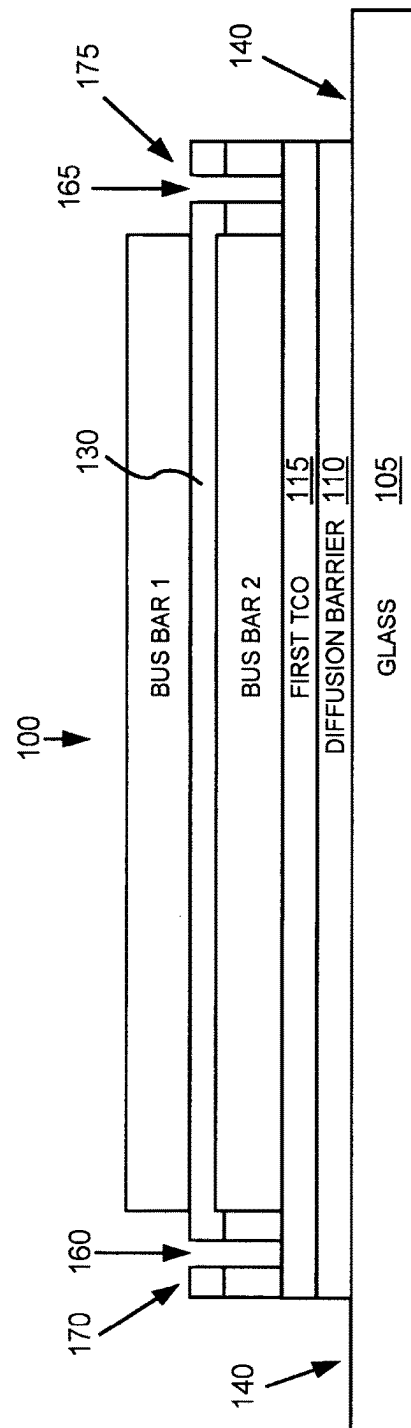
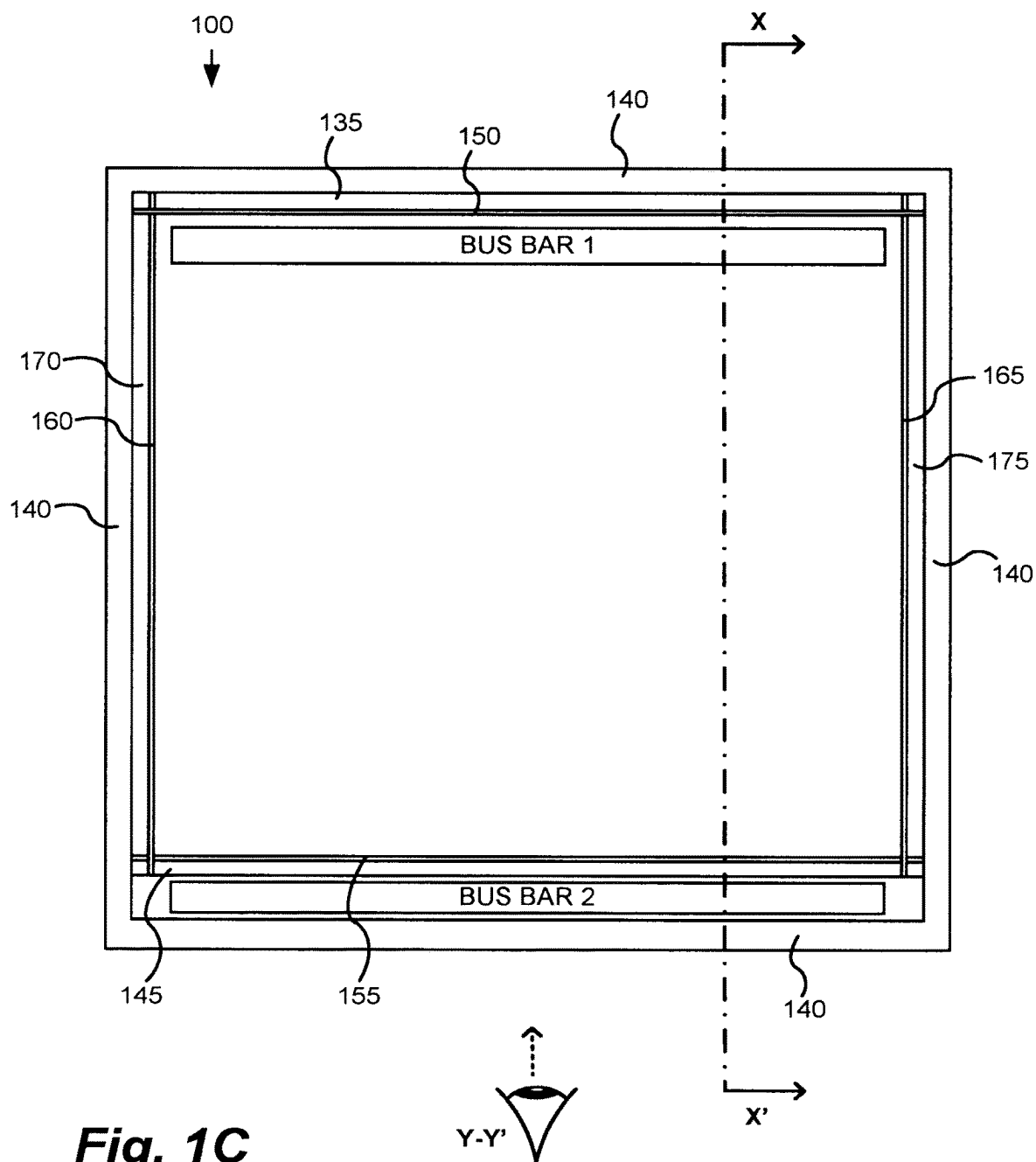


Fig. 1B



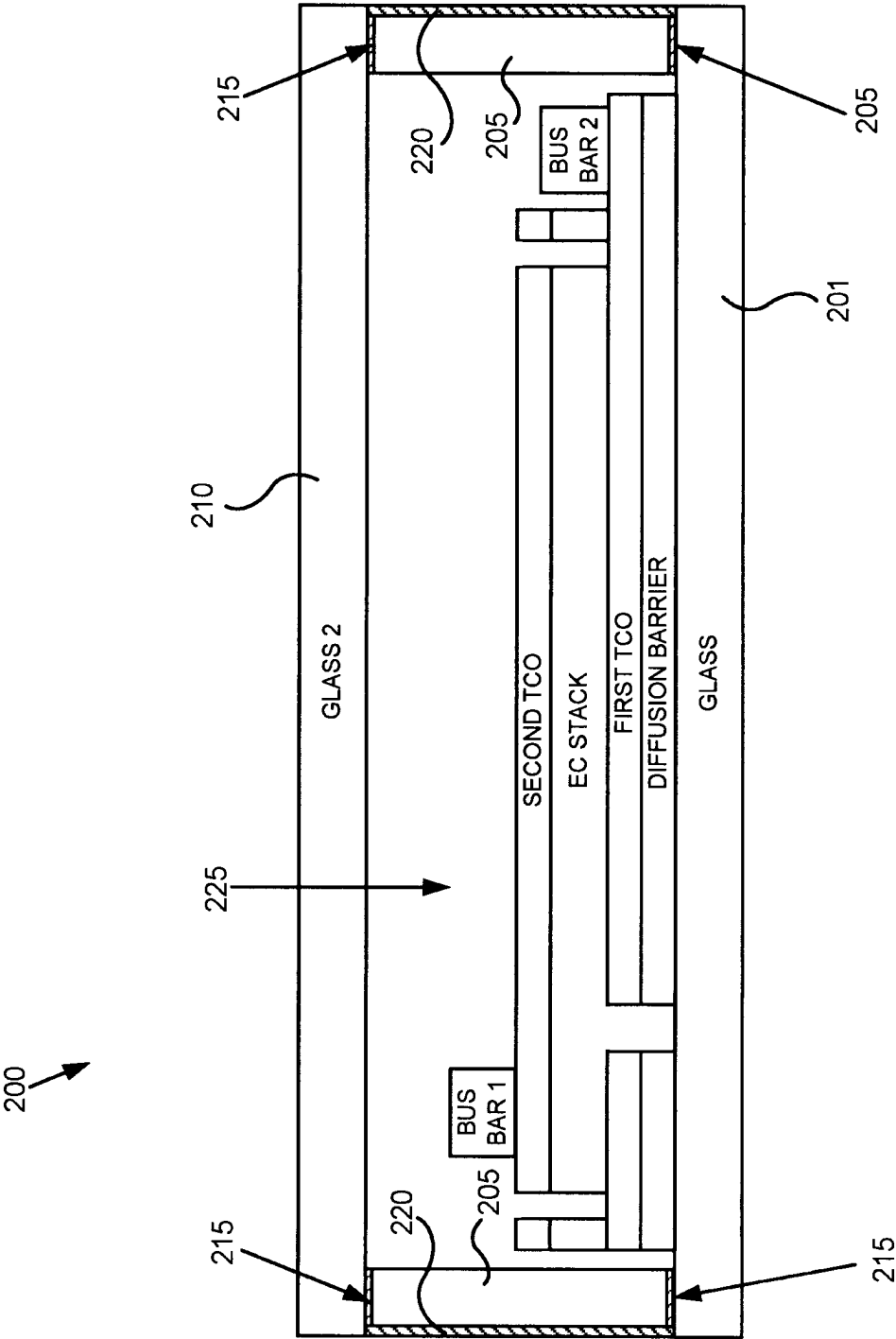


Fig. 2A

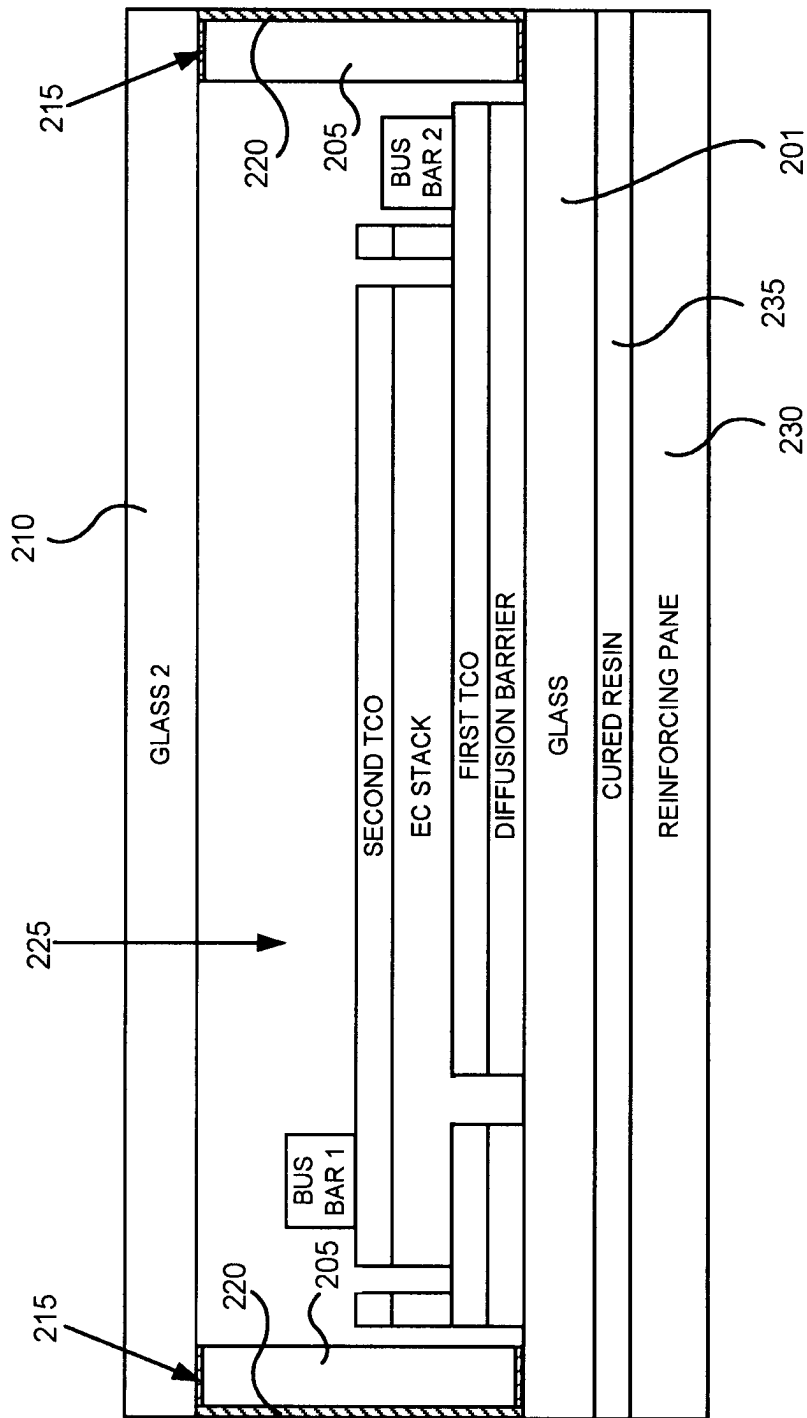
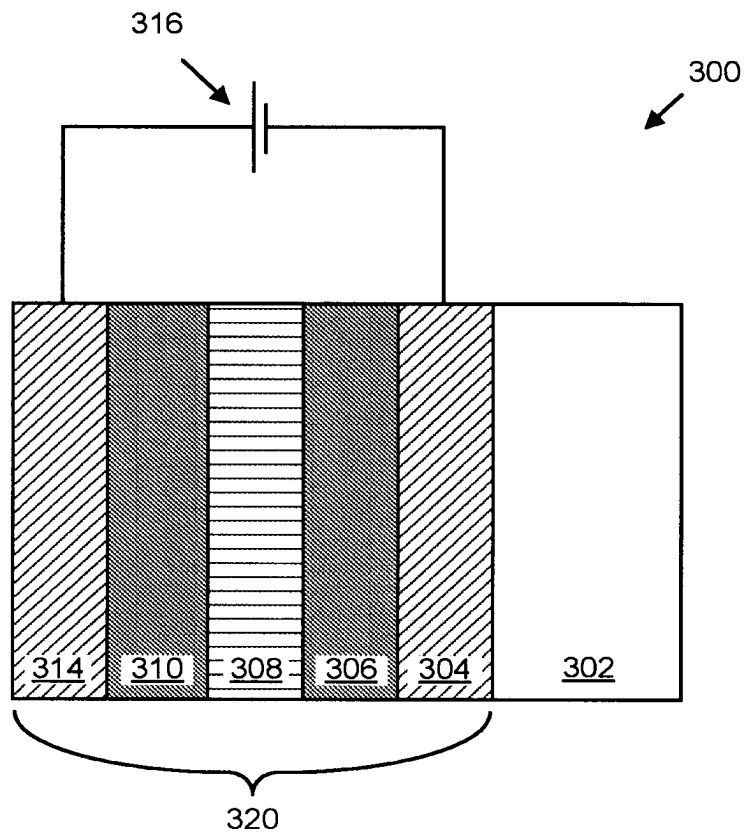
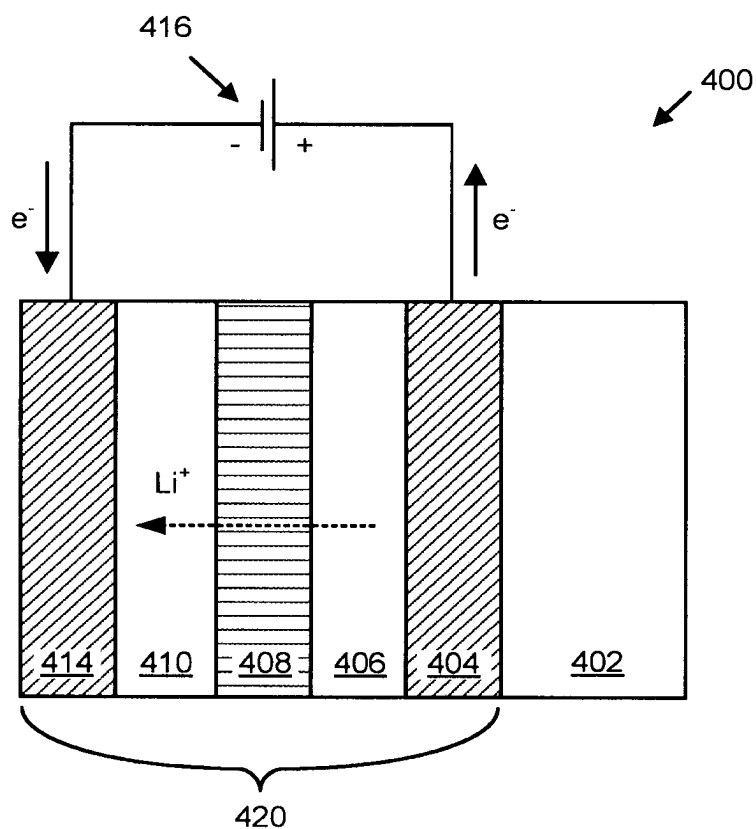
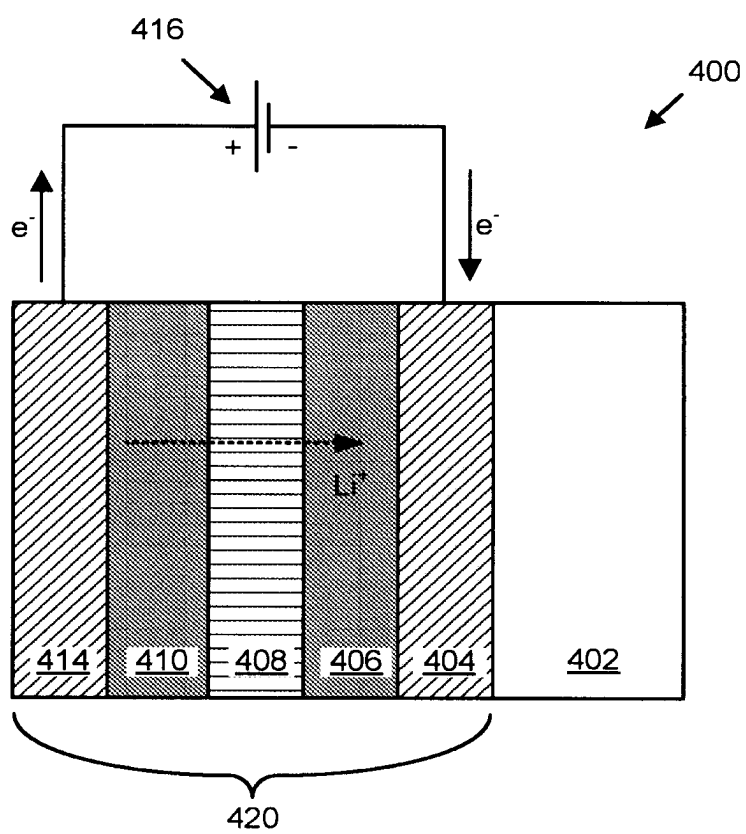
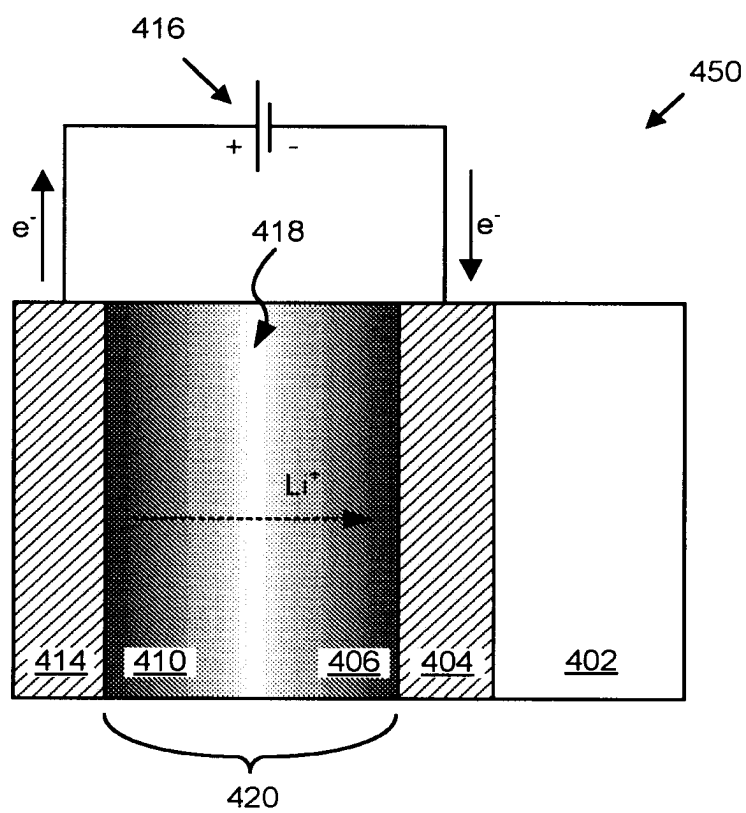


Fig. 2B

**FIG. 3A**

**FIG. 3B**

**FIG. 3C**

**FIG. 4**

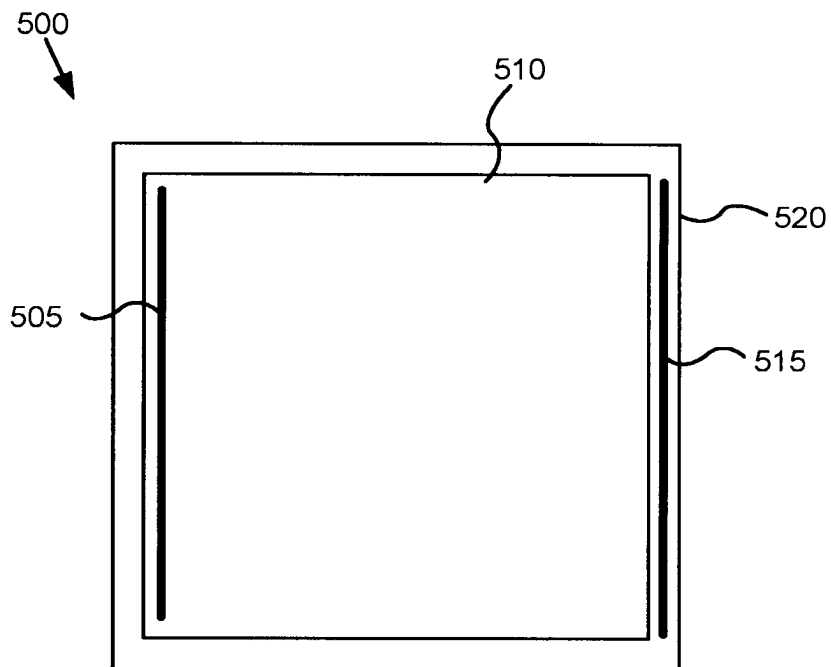


Fig. 5A

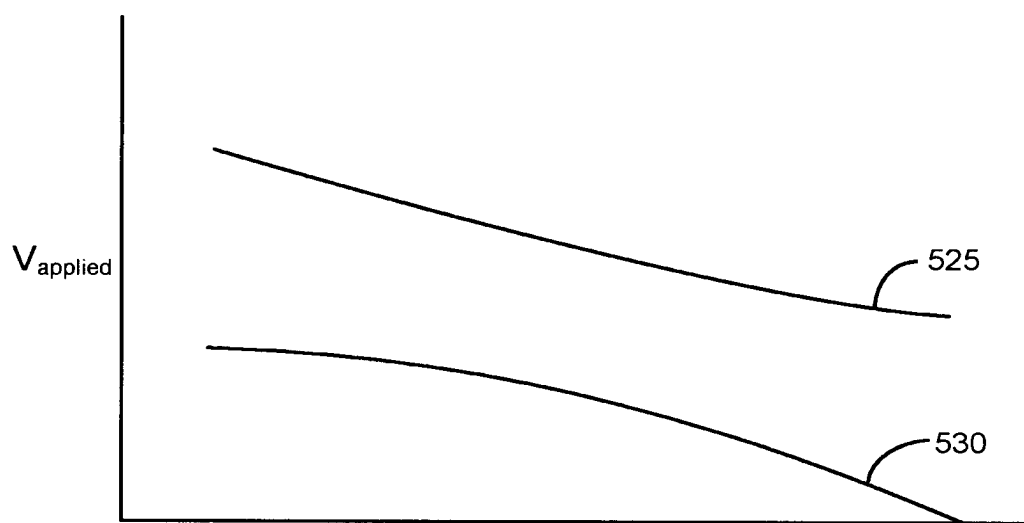


Fig. 5B

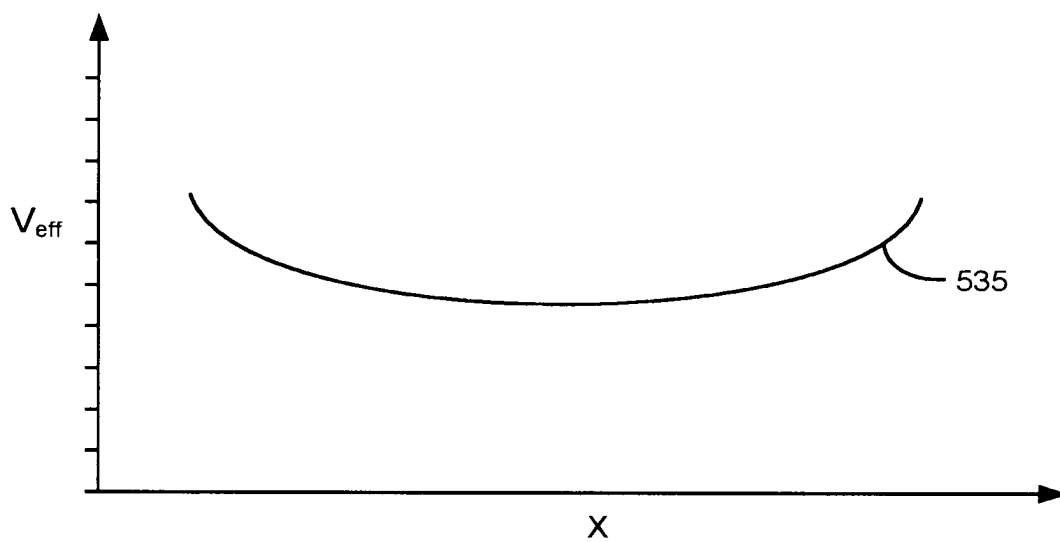


Fig. 5C

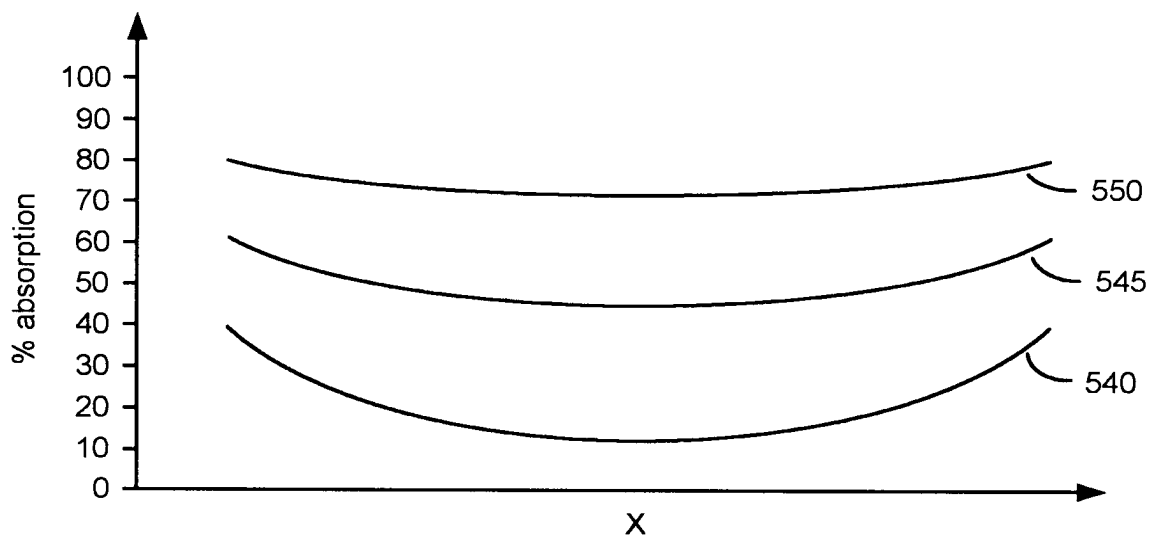


Fig. 5D

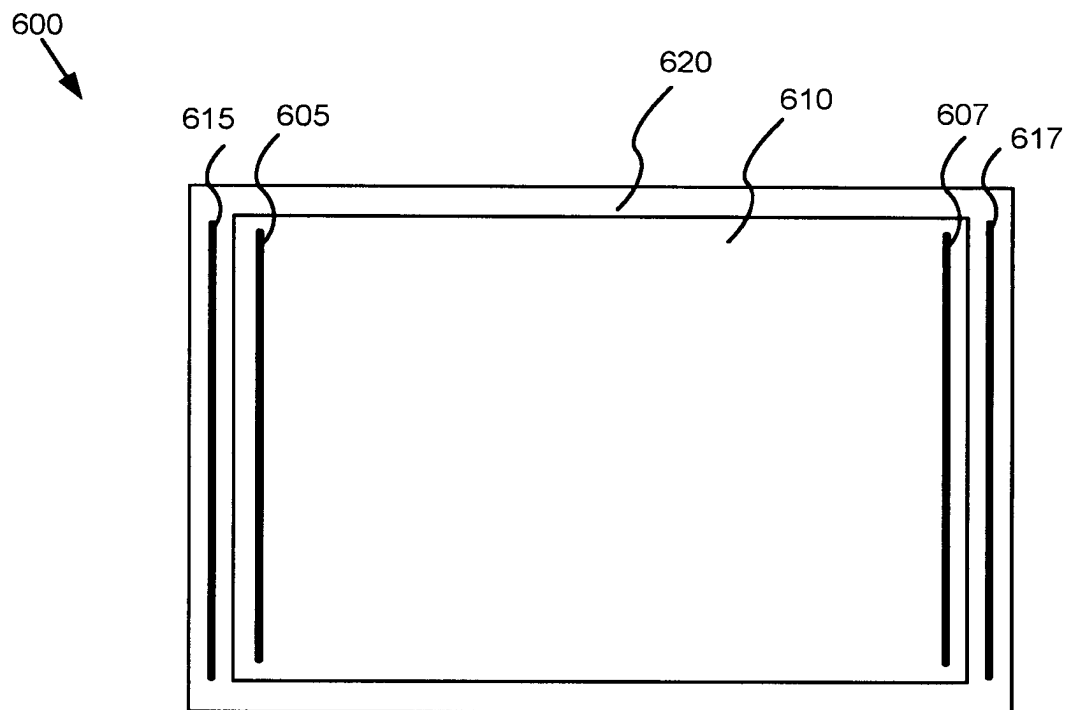


Fig. 6A

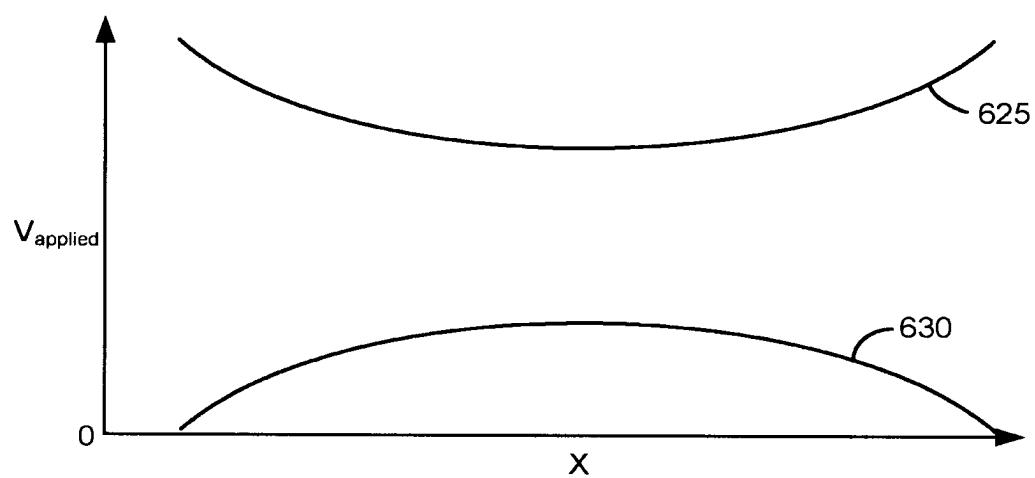


Fig. 6B

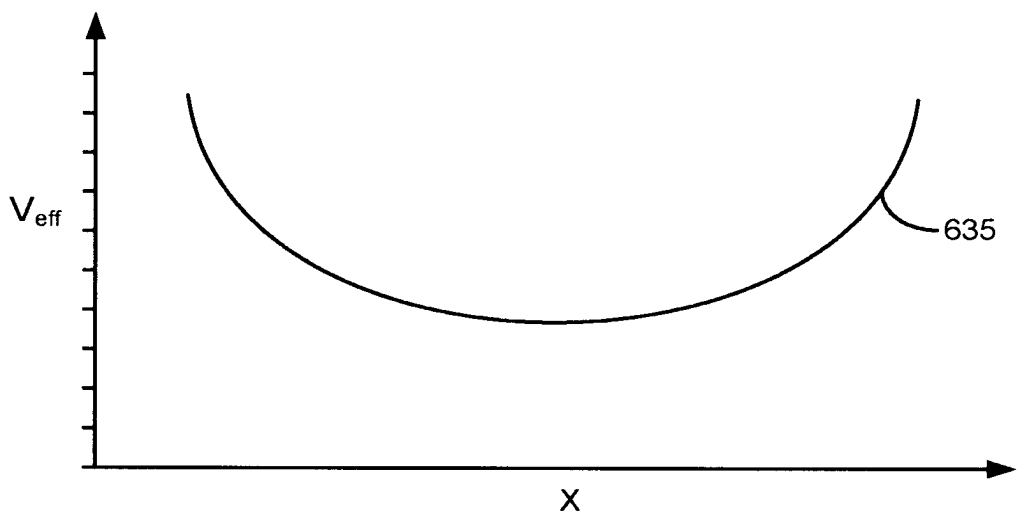


Fig. 6C

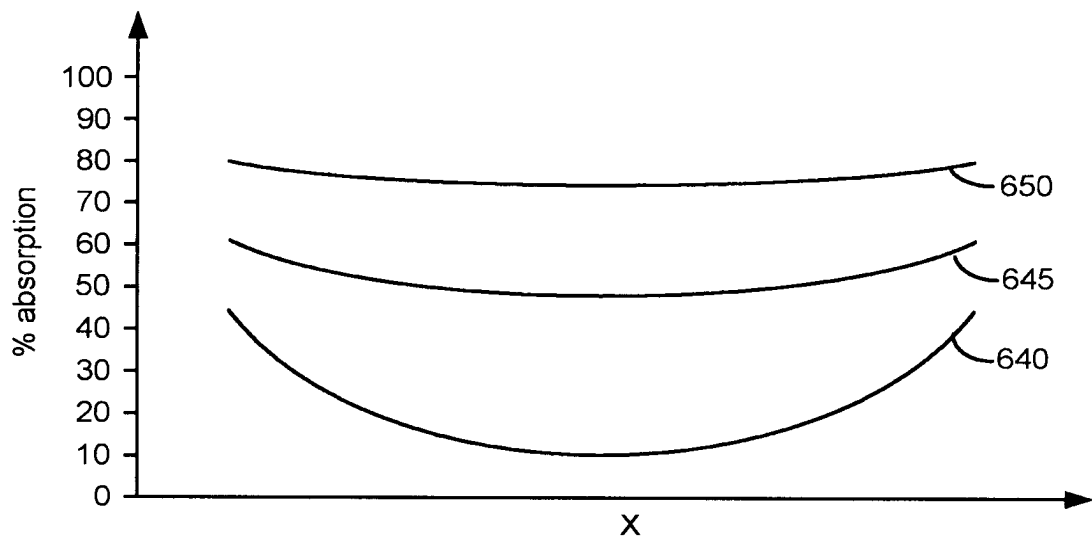


Fig. 6D

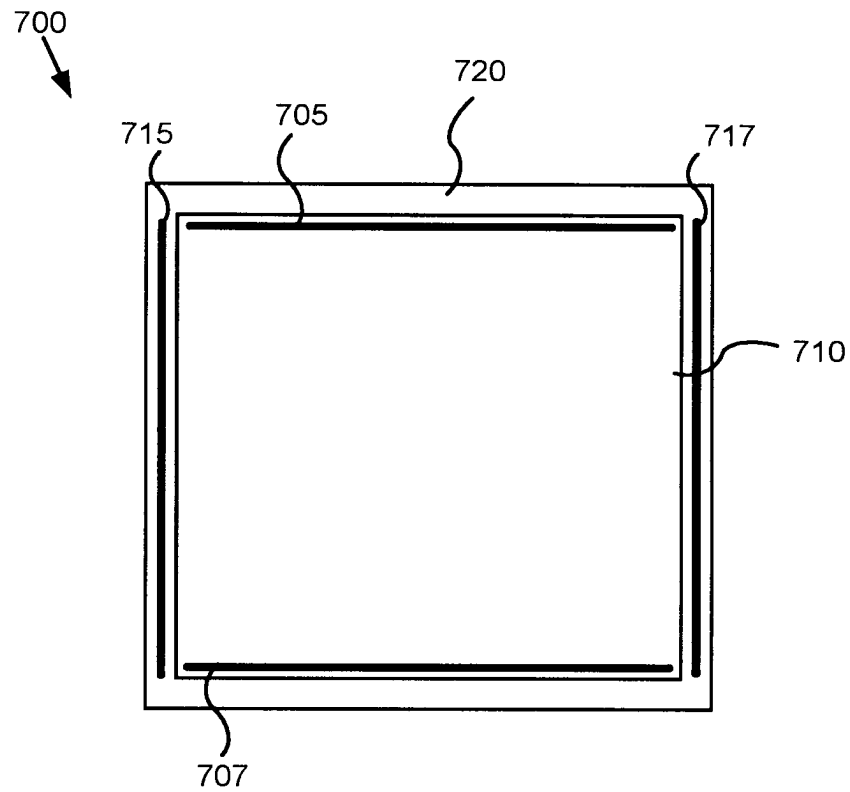


Fig. 7A

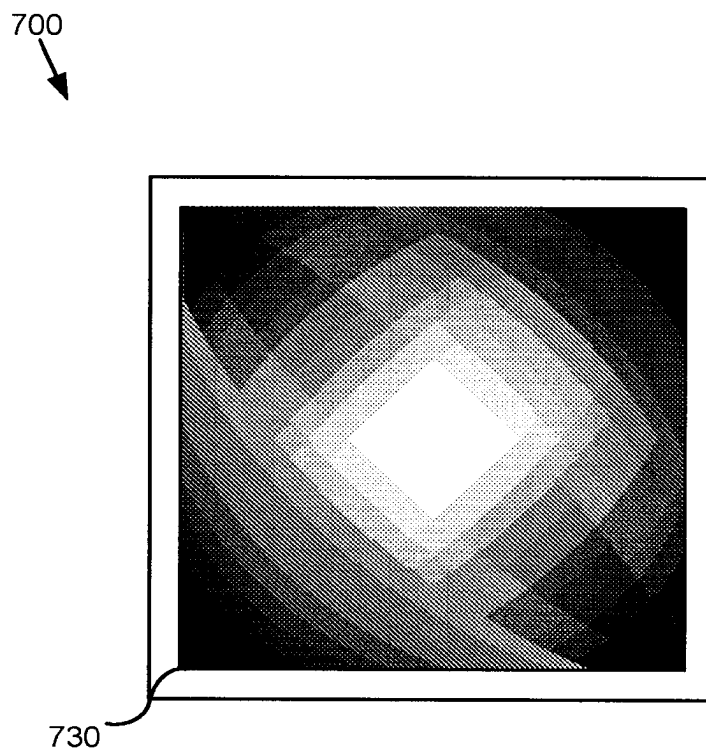


Fig. 7B

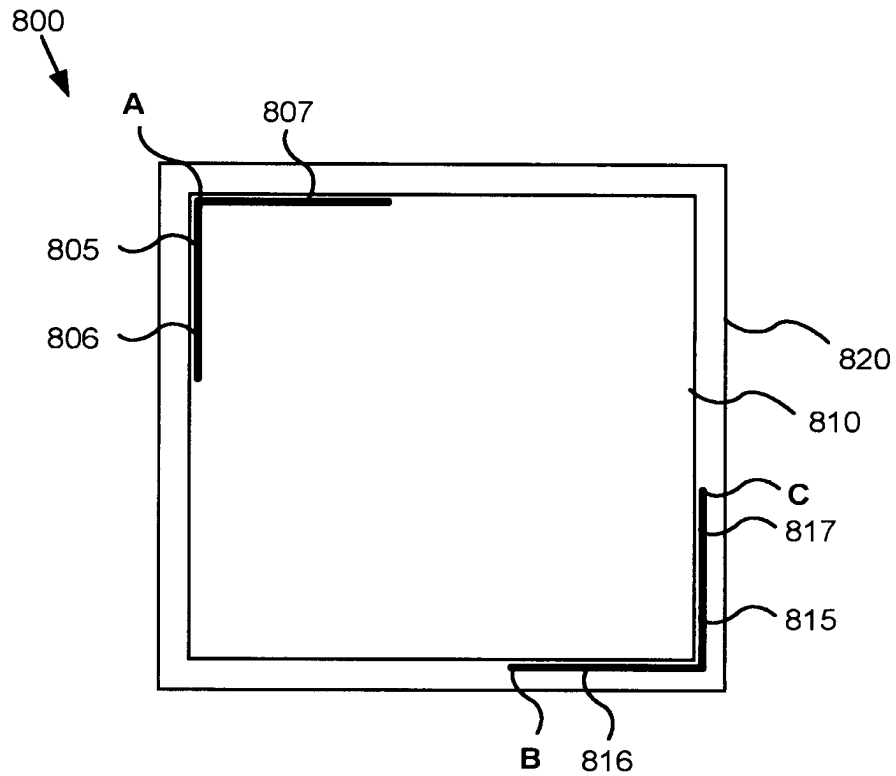


Fig. 8A

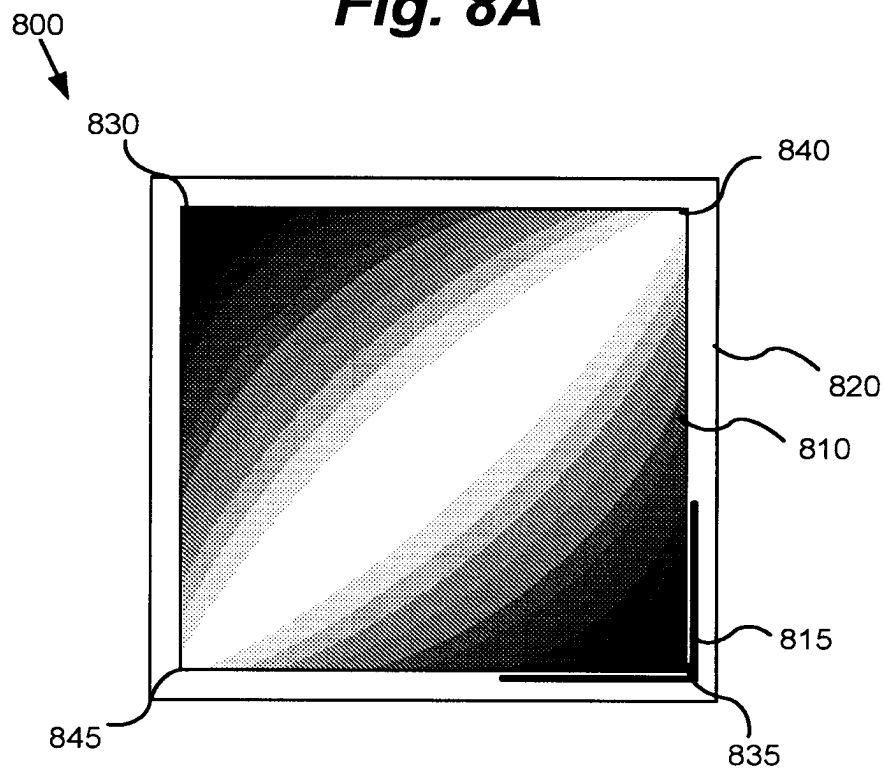


Fig. 8B

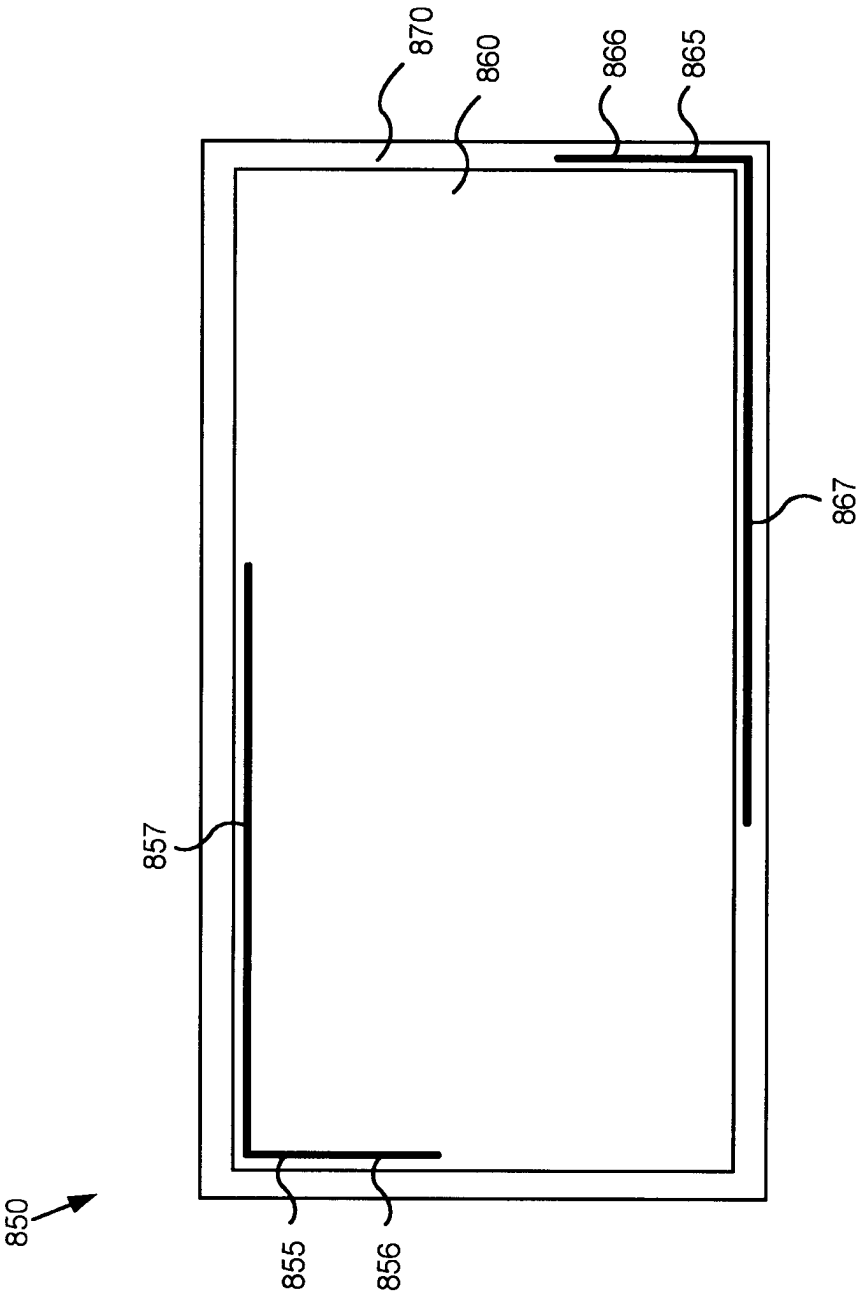
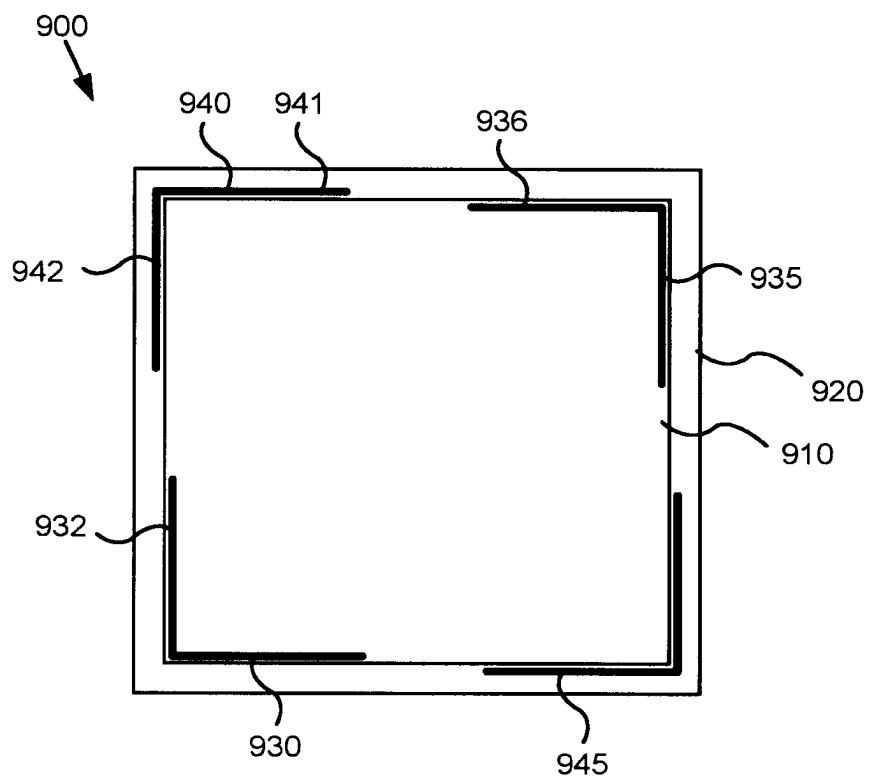


Fig. 8C

**Fig. 9**

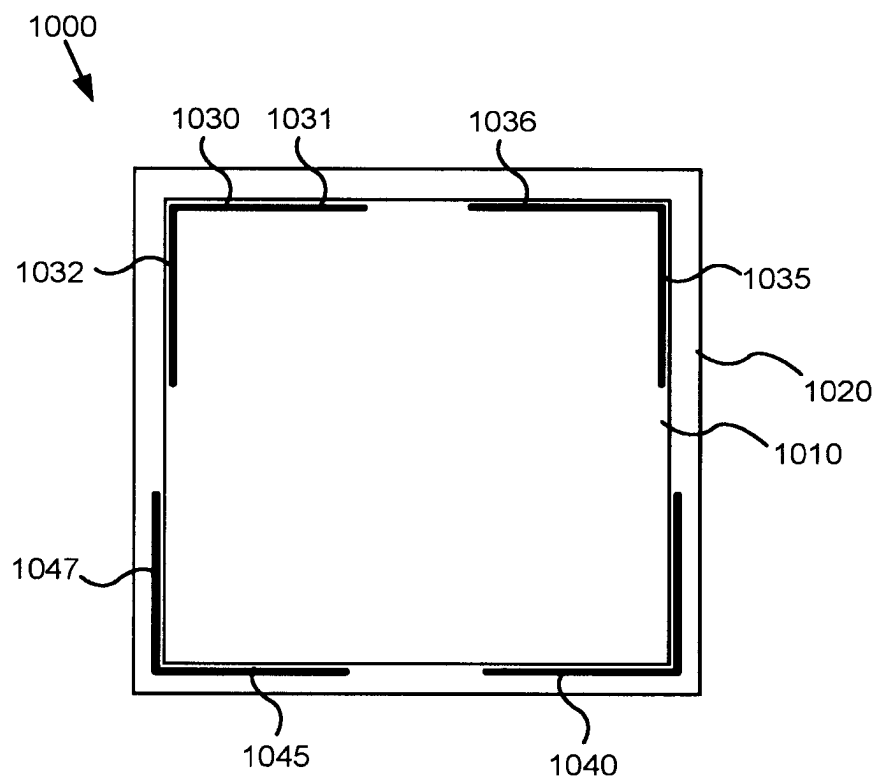


Fig. 10

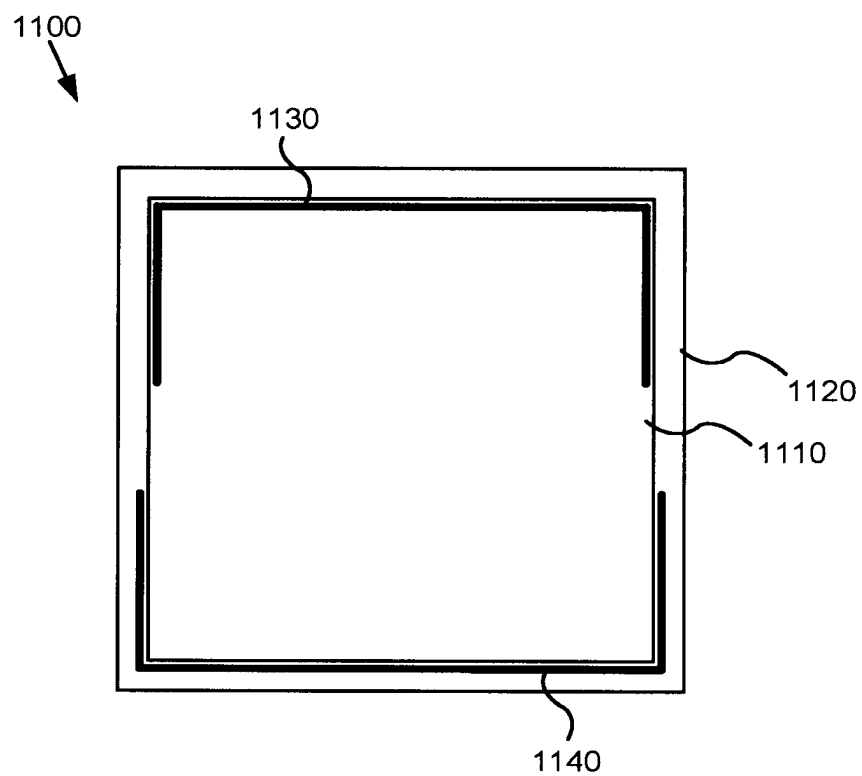


Fig. 11

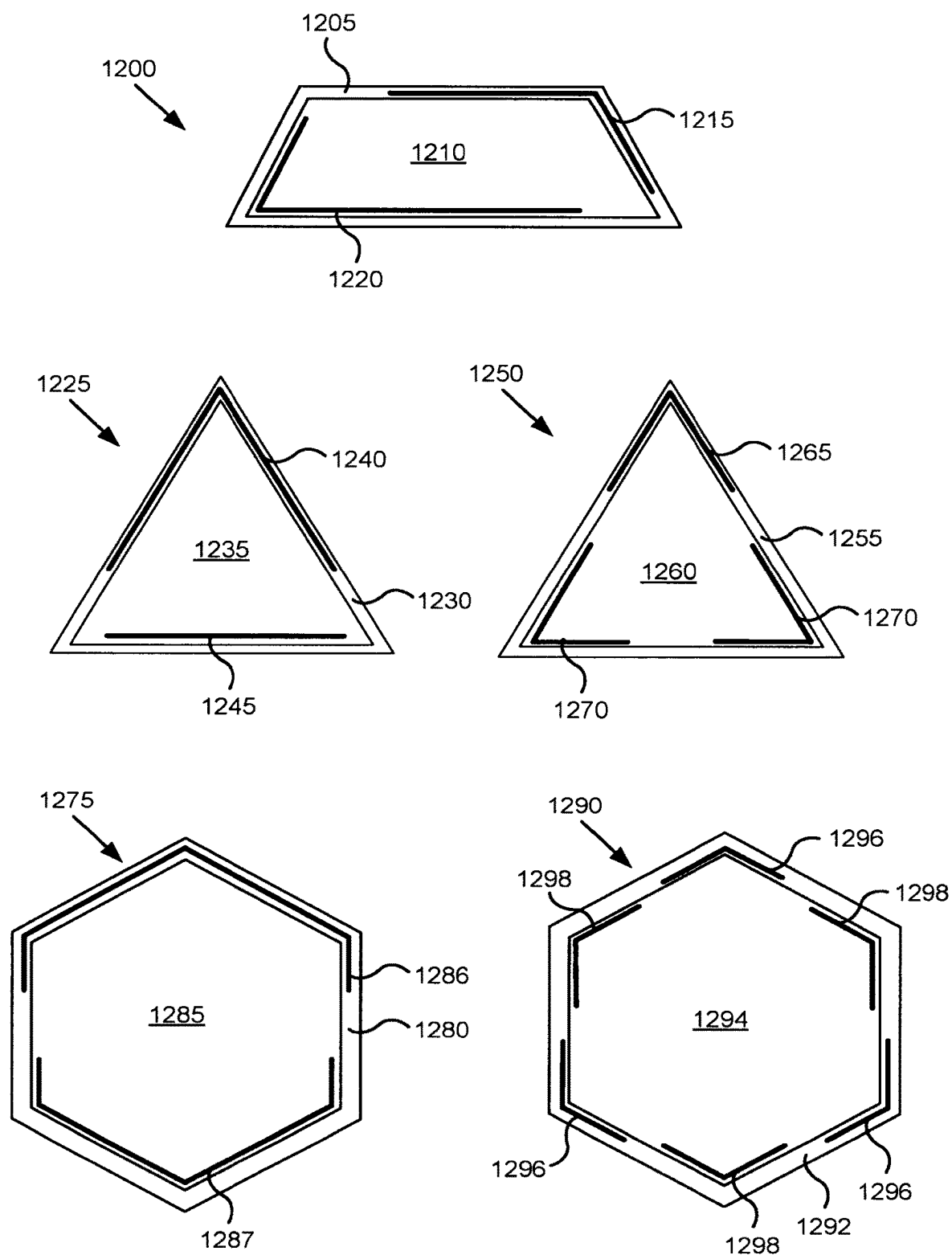
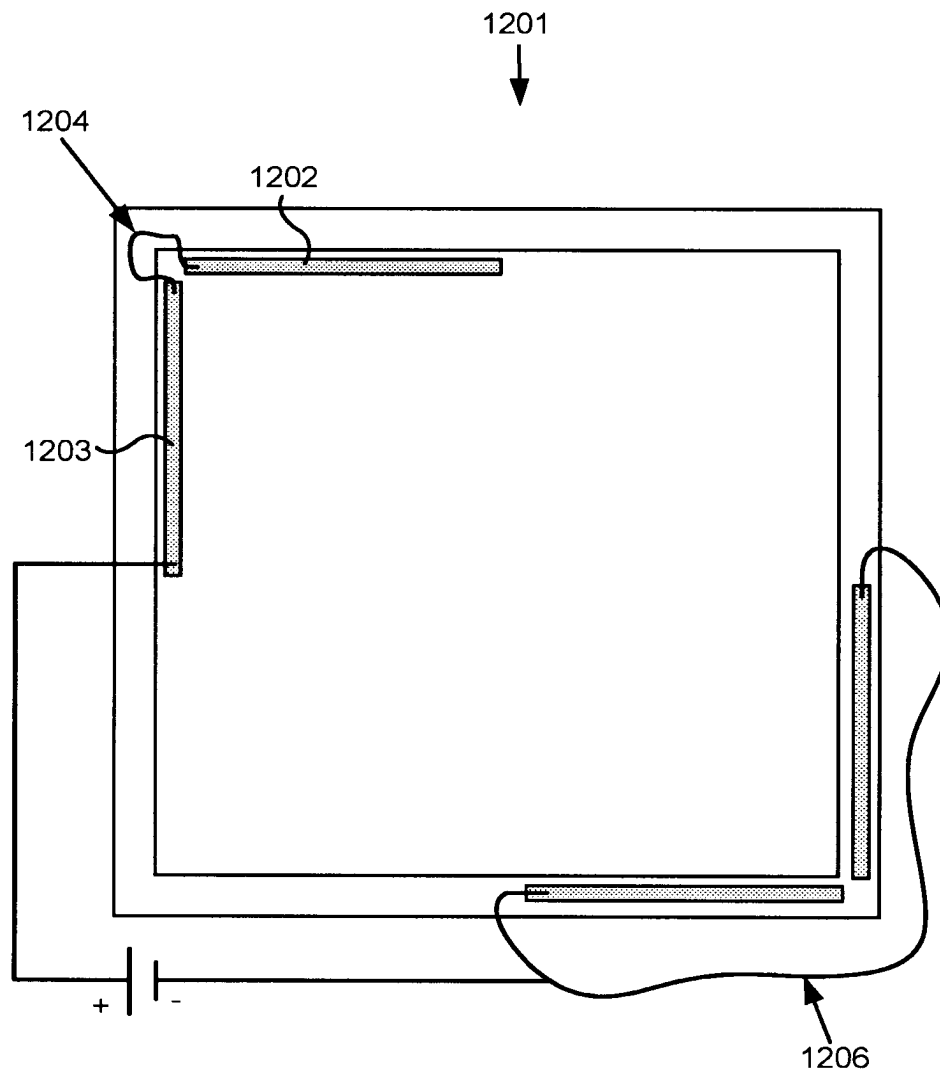
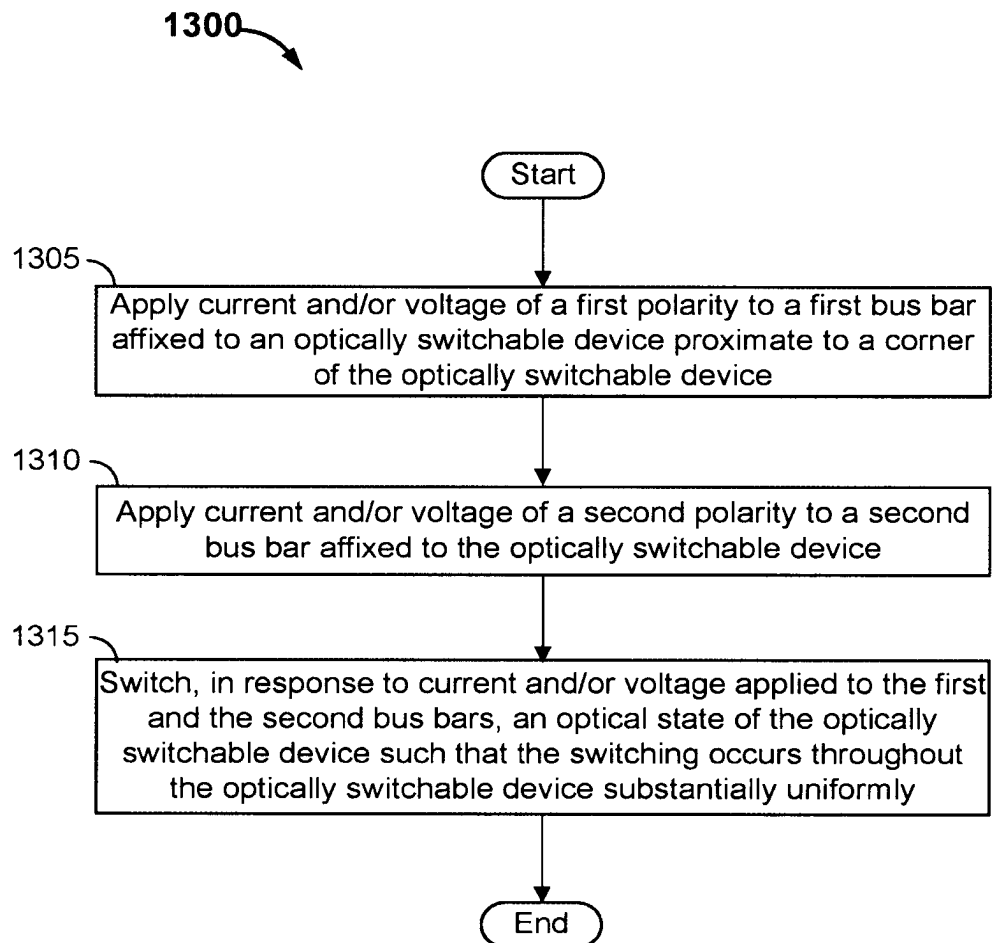
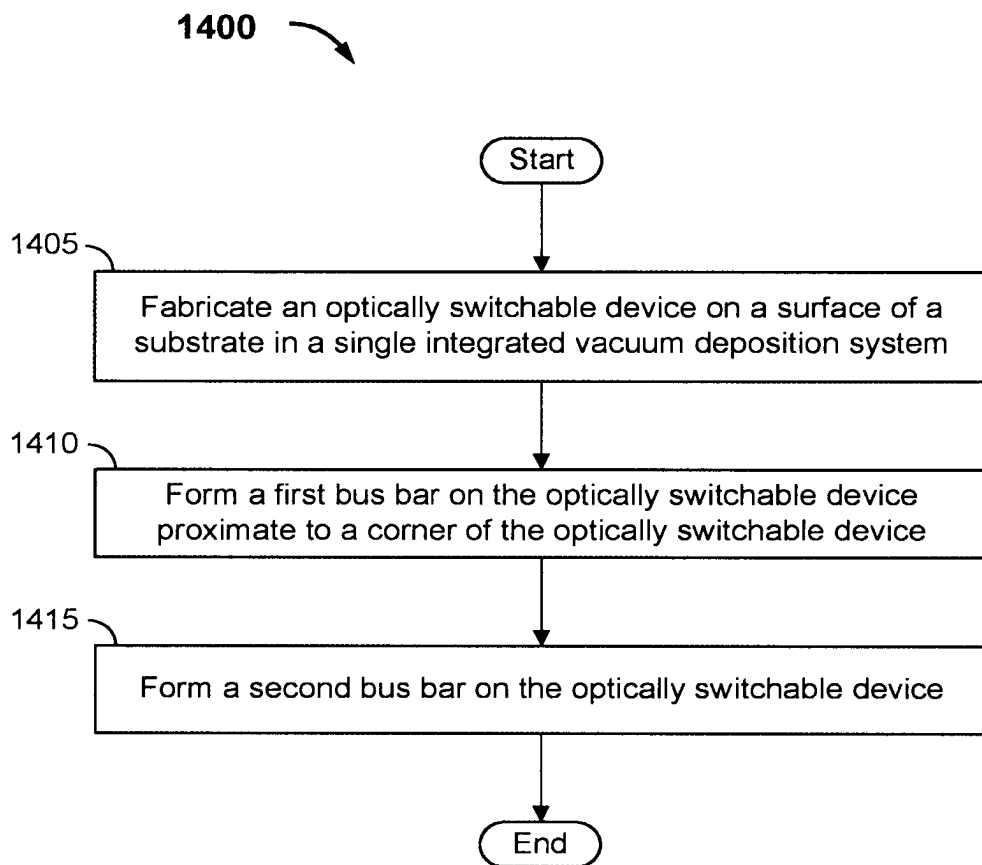
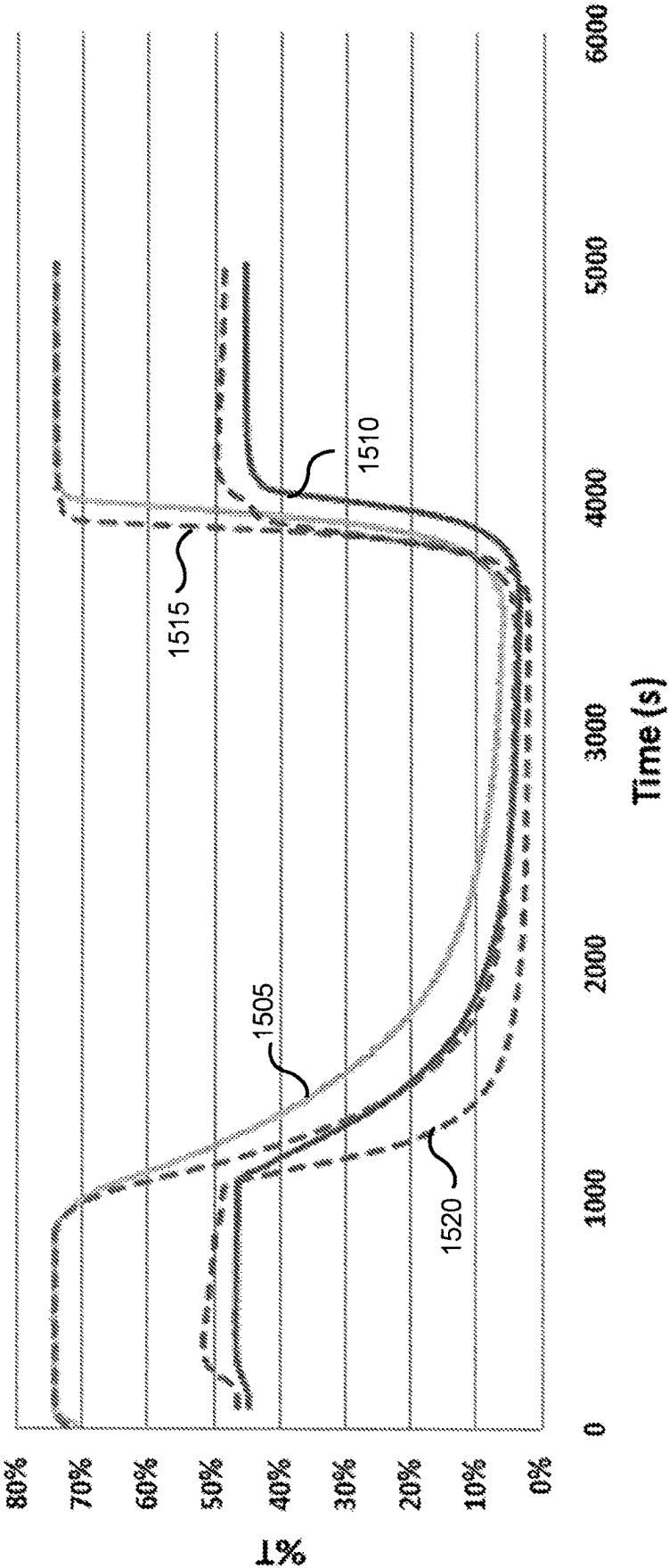


Fig. 12A

**Fig. 12B**

**Fig. 13**

**Fig. 14**



- 1505 planar bus bars, center
- 1510 planar bus bars, edge
- 1515 L-shaped bus bars, center
- 1520 L-shaped bus bars, edge

Fig. 15

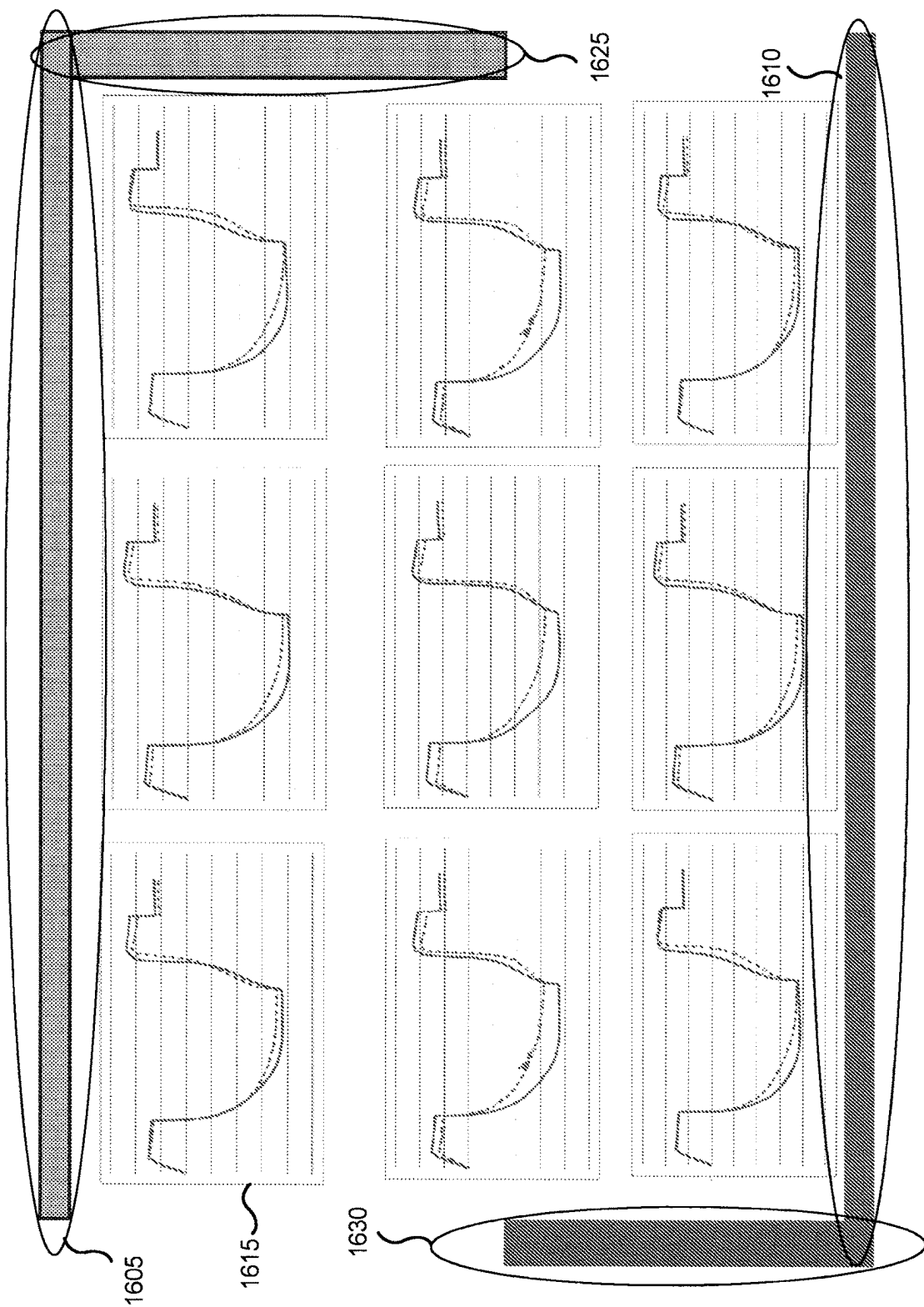
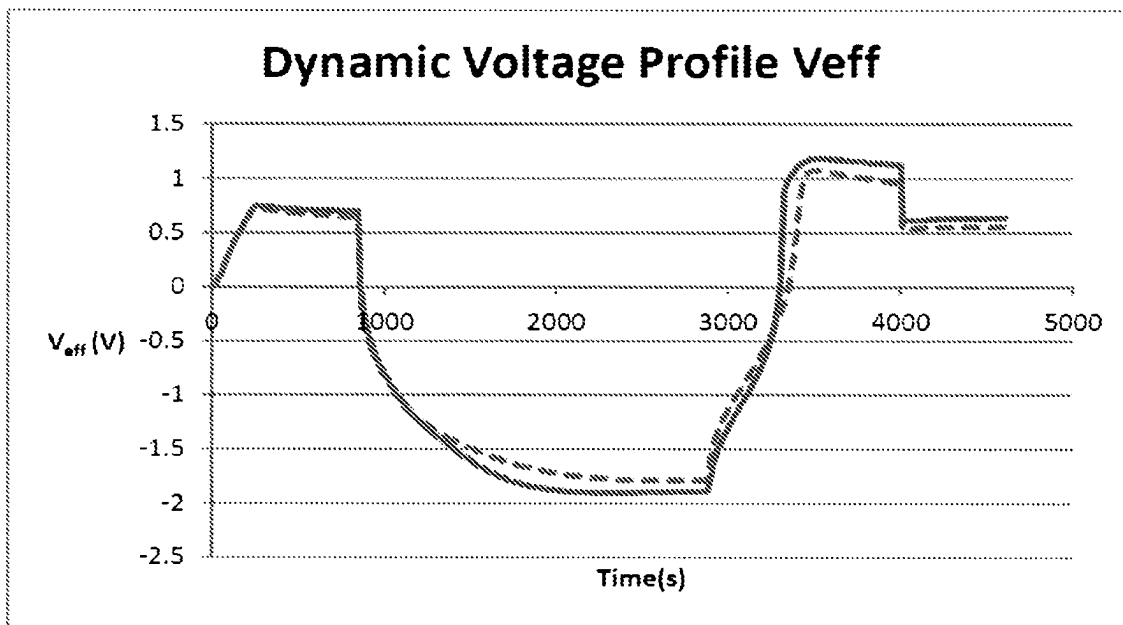


Fig. 16A

1615
↓**Fig. 16B**

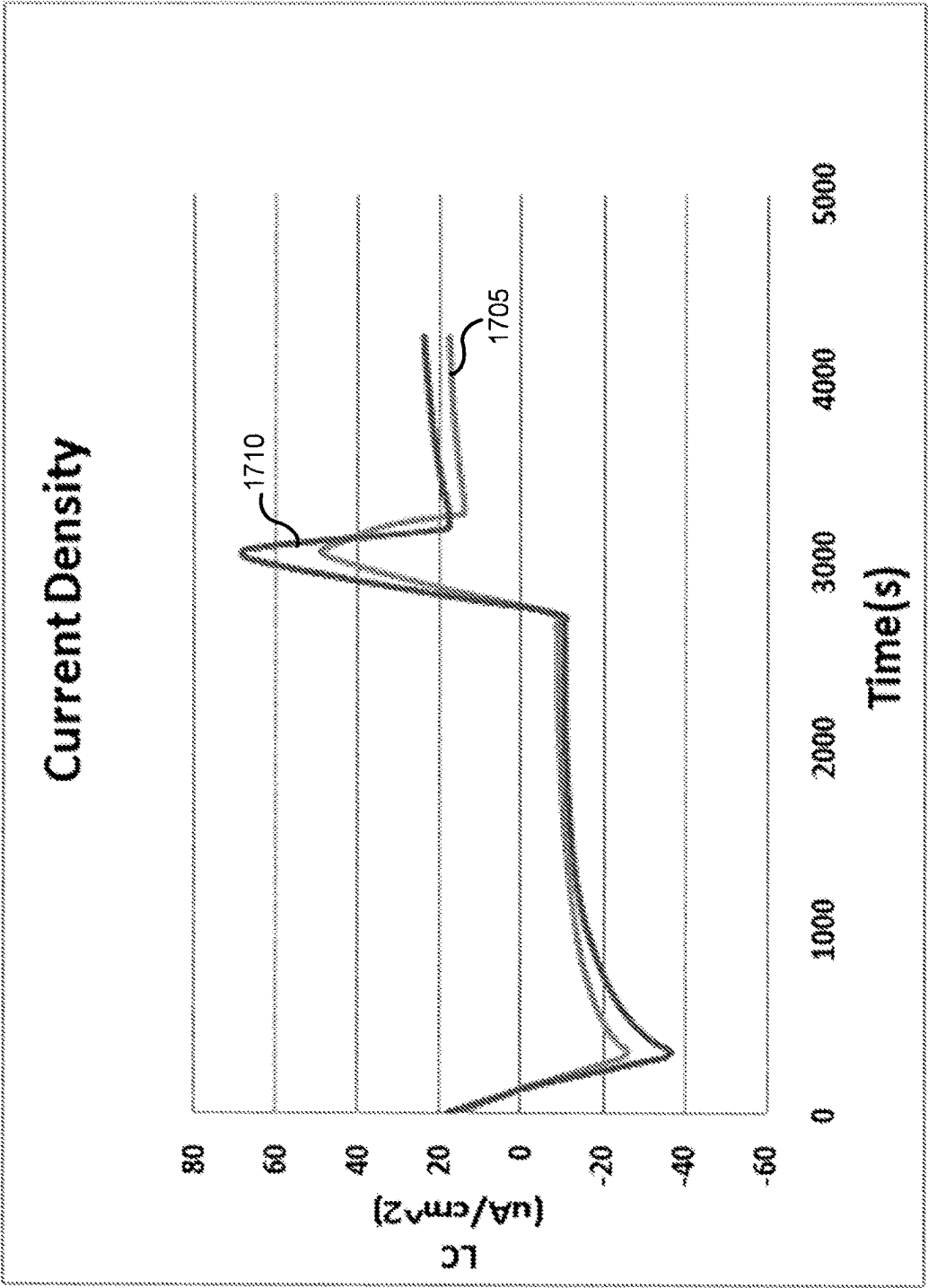


Fig. 17

1

ANGLED BUS BAR**CROSS REFERENCE TO RELATED APPLICATION**

An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in its entirety and for all purposes.

FIELD

The disclosed embodiments relate generally to bus bars for optically switchable devices, and more particularly to angled bus bars for thin film optically switchable devices.

BACKGROUND

Various optically switchable devices are available for controlling tinting, reflectivity, etc., of window panes or lites. Electrochromic devices are one example of optically switchable devices. Electrochromism is a phenomenon in which a material exhibits a reversible electrochemically-mediated change in an optical property when placed in a different electronic state, typically by being subjected to a voltage change. The optical property being manipulated is typically one or more of color, transmittance, absorbance, and reflectance. One well known electrochromic material is tungsten oxide (WO_3). Tungsten oxide is a cathodic electrochromic material in which a coloration transition, transparent to blue, occurs by electrochemical reduction.

Electrochromic materials may be incorporated into, for example, windows for home, commercial, and other uses. The color, transmittance, absorbance, and/or reflectance of such windows may be changed by inducing a change in the electrochromic material; i.e., electrochromic windows are windows that can be darkened or lightened electronically. A small voltage applied to an electrochromic device of the window will cause it to darken; reversing the voltage causes it to lighten. This capability allows for control of the amount of light that passes through the window, and presents an enormous opportunity for electrochromic windows to be used not only for aesthetic purposes but also for significant energy-savings. With energy conservation being foremost in modern energy policy, it is expected that growth of the electrochromic window industry will be robust in the coming years.

SUMMARY

Angled bus bars for thin film optically switchable devices are disclosed herein. Angled bus bars may allow for the transition of a thin film optically switchable device symmetrically and quickly, without overdriving the thin film optically switchable device.

In one embodiment, an apparatus includes a substrate with an optically switchable device disposed on a surface of the substrate. The optically switchable device has a perimeter with at least one corner including a first side, a first vertex joining the first side and a second side, and the second side. A first bus bar is affixed to the optically switchable device proximate to the corner and configured to deliver current and/or voltage for driving switching of the optically switchable device. The first bus bar includes a first arm and a second arm having a configuration that substantially

2

follows the shape of the first side, the first vertex, and the second side of the corner. A second bus bar is affixed to the optically switchable device and configured to deliver current and/or voltage for driving switching of the optically switchable device. The second bus bar may be angled or not. One embodiment is an electrochromic device including at least one angled bus bar as described herein.

In another embodiment, a method of changing an optical state of an optically switchable device includes applying current and/or voltage to at least one angled bus bar as described herein, which is in electrical communication with the optically switchable device. In one embodiment, the angled bus bar is as described above, and it is the first of two bus bars in electrical communication with the optically switchable device. The optically switchable device is disposed on a surface of a substrate and has a perimeter with at least one corner including a first side, a first vertex joining the first side and a second side, and the second side. The first bus bar is affixed to the optically switchable device proximate the corner. The first bus bar includes a first arm and a second arm having a configuration that substantially follows the shape of the first side, the first vertex, and the second side of the corner. A second bus bar, applied to an opposing electrode of the optically switchable device, may be angled or not. In response to current and/or voltage applied to the first and the second bus bars, an optical state of the optically switchable device changes in a substantially uniform manner.

In another embodiment, a method of fabricating an optically switchable device includes fabricating the optically switchable device on a surface of a substrate in a single integrated vacuum deposition system. The substrate is in a substantially vertical orientation in the integrated vacuum deposition system. The optically switchable device has a perimeter with at least one corner including a first side, a first vertex joining the first side and a second side, and the second side. A first bus bar is formed on the optically switchable device proximate to the corner. The first bus bar includes a first arm and a second arm having a configuration that substantially follows the shape of the first side, the first vertex, and the second side of the corner. The first bus bar is configured to deliver current and/or voltage for driving switching of the optically switchable device.

In some embodiments, the method further includes forming a second bus bar on the optically switchable device. The second bus bar is configured to deliver current and/or voltage for driving switching of the optically switchable device. In some embodiments, the perimeter of the optically switchable device has a second corner including a third side, a second vertex joining the third side and a fourth side, and the fourth side. The second bus bar includes a third arm and a fourth arm having a configuration that substantially follows the shape of the third side, the second vertex, and the fourth side of the second corner.

In another embodiment, an electrochromic lite includes a substantially transparent substrate with an electrochromic device thereon. The electrochromic device has an area bounded by four sides. The electrochromic lite further includes a first bus bar and a second bus bar. The first bus bar is in electrical communication with a bottom transparent conductor of the electrochromic device. The second bus bar is in electrical communication with a top transparent conductor of the electrochromic device. The first bus bar is diagonally opposed to the second bus bar. Each of the first and second bus bars spans diagonally opposing vertices of the area and at least some portion of the two sides forming each of the diagonally opposing vertices.

3

In some embodiments, the area of the electrochromic device is a substantially rectangular area. The at least some portion spanned by the first and the second bus bars is between about 10% and about 90%, between about 10% and about 65%, or between about 35% and about 65% of the length of the respective sides of the substantially rectangular area.

In some embodiments, the area of the electrochromic device is a substantially rectangular area. Each of the first and second bus bars is L-shaped. A first arm of each of the first and second bus bars traverses a portion of a longer side of the substantially rectangular area. A second arm of each of the first and second bus bars traverses a portion of a shorter side, substantially orthogonal to the longer side, of the substantially rectangular area. The first arm is longer than the second arm. In some embodiments, the first arm spans between about 75% and about 90% of the longer side, and the second arm spans between about 25% and about 75% of the shorter side. In some other embodiments, the substantially rectangular area is square and all the arms of the first and second bus bars are of substantially equal length.

In some embodiments, the first bus bar and the second bus bar are configured such that the optically switchable device switches from a first optical state to a second optical state in about 10 minutes or less. In some embodiments, the electrochromic device is all solid state and inorganic.

These and other features and advantages will be described in further detail below, with reference to the associated drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1C show schematic diagrams of electrochromic devices formed on glass substrates, i.e. electrochromic lites.

FIGS. 2A and 2B show cross-sectional schematic diagrams of the electrochromic lites as described in relation to FIGS. 1A-C integrated into an IGU.

FIG. 3A is a schematic cross-section of an electrochromic device.

FIG. 3B is a schematic cross-section of an electrochromic device in a bleached state (or transitioning to a bleached state).

FIG. 3C is a schematic cross-section of the electrochromic device shown in FIG. 3B, but in a colored state (or transitioning to a colored state).

FIG. 4 is a schematic cross-section of an electrochromic device in a colored state, where the device has an interfacial region which does not contain a distinct ion conductor layer.

FIG. 5A shows a schematic diagram of an electrochromic lite with planar bus bars.

FIGS. 5B-5D show diagrams associated with the operation of the electrochromic lite shown in FIG. 5A.

FIG. 6A shows a schematic diagram of an electrochromic lite with cylindrical bus bars.

FIGS. 6B-6D show diagrams associated with the operation of the electrochromic lite shown in FIG. 6A.

FIG. 7A shows a schematic diagram of an electrochromic lite with perpendicular cylindrical bus bars.

FIG. 7B shows a diagram associated with the operation of the electrochromic lite shown in FIG. 7A.

FIG. 8A shows a schematic diagram of a square-shaped electrochromic lite with angled bus bars.

FIG. 8B shows a diagram associated with the operation of the electrochromic lite shown in FIG. 8A.

FIG. 8C shows a schematic diagram of a rectangle-shaped electrochromic lite with angled bus bars.

4

FIGS. 9-11 shows schematic diagrams of electrochromic lites including angled bus bars in different configurations.

FIG. 12A shows schematic diagrams of electrochromic lites of varying shapes including angled bus bars in different configurations.

FIG. 12B shows schematic diagrams of angled bus bars in alternative configurations.

FIG. 13 shows a flowchart depicting a method of changing an optical state of an optically switchable device.

FIG. 14 shows a flowchart depicting a method of fabricating an optically switchable device.

FIG. 15 shows a graph including plots of the percent transmittance of an electrochromic lite with planar bus bars and an electrochromic lite with angled bus bars.

FIG. 16A shows a graph including plots of the voltage at different positions on a lower conductive electrode of an electrochromic lite.

FIG. 16B shows an enlarged view of the plot from the upper left hand corner of FIG. 16A.

FIG. 17 shows a graph including plots of the current density in an electrochromic lite with planar bus bars and an electrochromic lite with angled bus bars.

DETAILED DESCRIPTION

Introduction and Overview of Electrochromic Devices: Fabrication

It should be understood that while the disclosed embodiments focus on electrochromic (EC) windows (also referred to as smart windows), the concepts disclosed herein may apply to other types of switchable optical devices, including liquid crystal devices, suspended particle devices, and the like. For example, a liquid crystal device or a suspended particle device, instead of an electrochromic device, could be incorporated in any of the disclosed embodiments.

In order to orient the reader to the embodiments of angled bus bars disclosed herein, a brief discussion of electrochromic devices is provided. This initial discussion of electrochromic devices is provided for context only, and the subsequently described embodiments of angled bus bars are not limited to the specific features and fabrication processes of this initial discussion.

A particular example of an electrochromic lite is described with reference to FIGS. 1A-1C, in order to illustrate embodiments described herein. FIG. 1A is a cross-sectional representation (see cut X-X' of FIG. 1C) of an electrochromic lite, **100**, which is fabricated starting with a glass sheet, **105**. FIG. 1B shows an end view (see perspective Y-Y' of FIG. 1C) of EC lite **100**, and FIG. 1C shows a top-down view of EC lite **100**. FIG. 1A shows the electrochromic lite after fabrication on glass sheet **105**, edge deleted to produce area, **140**, around the perimeter of the lite. The electrochromic lite has also been laser scribed and bus bars have been attached. The glass lite **105** has a diffusion barrier, **110**, and a first transparent conducting oxide (TCO), **115**, on the diffusion barrier. In this example, the edge deletion process removes both TCO **115** and diffusion barrier **110**, but in other embodiments only the TCO is removed, leaving the diffusion barrier intact. The TCO **115** is the first of two conductive layers used to form the electrodes of the electrochromic device fabricated on the glass sheet. In this example, the glass sheet includes underlying glass and the diffusion barrier layer. Thus, in this example, the diffusion barrier is formed, and then the first TCO, an EC stack, **125**, (e.g., having electrochromic, ion conductor, and counter electrode layers), and a second TCO, **130**, are formed. In one embodiment, the electrochromic device (EC stack and sec-

5

ond TCO) is fabricated in an integrated deposition system where the glass sheet does not leave the integrated deposition system at any time during fabrication of the stack. In one embodiment, the first TCO layer is also formed using the integrated deposition system where the glass sheet does not leave the integrated deposition system during deposition of the EC stack and the (second) TCO layer. In one embodiment, all of the layers (diffusion barrier, first TCO, EC stack, and second TCO) are deposited in the integrated deposition system where the glass sheet does not leave the integrated deposition system during deposition. In this example, prior to deposition of EC stack **125**, an isolation trench, **120**, is cut through TCO **115** and diffusion barrier **110**. Trench **120** is made in contemplation of electrically isolating an area of TCO **115** that will reside under bus bar **1** after fabrication is complete (see FIG. 1A). This is done to avoid charge buildup and coloration of the EC device under the bus bar, which can be undesirable.

After formation of the EC device, edge deletion processes and additional laser scribing are performed. FIG. 1A depicts areas **140** where the device has been removed, in this example, from a perimeter region surrounding laser scribe trenches, **150**, **155**, **160**, and **165**, which pass through second TCO **130** and the EC stack, but not the first TCO **115**. Laser scribe trenches **150**, **155**, **160**, and **165** are made to isolate portions of the EC device, **135**, **145**, **170**, and **175**, which were potentially damaged during edge deletion processes from the operable EC device. In one embodiment, laser scribe trenches **150**, **160**, and **165** pass through the first TCO to aid in isolation of the device (laser scribe trench **155** does not pass through the first TCO, otherwise it would cut off bus bar **2**'s electrical communication with the first TCO and thus the EC stack). The laser or lasers used for the laser scribe processes are typically, but not necessarily, pulse-type lasers, for example, diode-pumped solid state lasers. For example, the laser scribe processes can be performed using a suitable laser from IPG Photonics (of Oxford, Massachusetts), or from Ekspla (of Vilnius, Lithuania). Scribing can also be performed mechanically, for example, by a diamond tipped scribe. One of ordinary skill in the art would appreciate that the laser scribing processes can be performed at different depths and/or performed in a single process whereby the laser cutting depth is varied, or not, during a continuous path around the perimeter of the EC device. In one embodiment, the edge deletion is performed to the depth of the first TCO.

After laser scribing is complete, bus bars are attached. Non-penetrating bus bar (**1**) is applied to the second TCO. Non-penetrating bus bar (**2**) is applied to an area where the device was not deposited (e.g., from a mask protecting the first TCO from device deposition), in contact with the first TCO or, in this example, where an edge deletion process (e.g., laser ablation using an apparatus having a XY or XYZ galvanometer) was used to remove material down to the first TCO. In this example, both bus bar **1** and bus bar **2** are non-penetrating bus bars. A penetrating bus bar is one that is typically pressed into and through the EC stack to make contact with the TCO at the bottom of the stack. A non-penetrating bus bar is one that does not penetrate into the EC stack layers, but rather makes electrical and physical contact on the surface of a conductive layer, for example, a TCO.

The TCO layers can be electrically connected using a non-traditional bus bar, for example, a bus bar fabricated with screen and lithography patterning methods. In one embodiment, electrical communication is established with the device's transparent conducting layers via silk screening (or using another patterning method) a conductive ink

6

followed by heat curing or sintering the ink. Advantages to using the above described device configuration include simpler manufacturing, for example, and less laser scribing than conventional techniques which use penetrating bus bars.

After the bus bars are connected, the device is integrated into an insulated glass unit (IGU), which includes, for example, wiring the bus bars and the like. In some embodiments, one or both of the bus bars are inside the finished IGU, however in one embodiment one bus bar is outside the seal of the IGU and one bus bar is inside the IGU. In the former embodiment, area **140** is used to make the seal with one face of the spacer used to form the IGU. Thus, the wires or other connection to the bus bars runs between the spacer and the glass. As many spacers are made of metal, e.g., stainless steel, which is conductive, it is desirable to take steps to avoid short circuiting due to electrical communication between the bus bar and connector thereto and the metal spacer.

As described above, after the bus bars are connected, the electrochromic lite is integrated into an IGU, which includes, for example, wiring for the bus bars and the like. In the embodiments described herein, both of the bus bars are inside the primary seal of the finished IGU. FIG. 2A shows a cross-sectional schematic diagram of the electrochromic window as described in relation to FIGS. 1A-C integrated into an IGU, **200**. A spacer, **205**, is used to separate the electrochromic lite from a second lite, **210**. Second lite **210** in IGU **200** is a non-electrochromic lite, however, the embodiments disclosed herein are not so limited. For example, lite **210** can have an electrochromic device thereon and/or one or more coatings such as low-E coatings and the like. Lite **201** can also be laminated glass, such as depicted in FIG. 2B (lite **201** is laminated to reinforcing pane, **230**, via resin, **235**). Between spacer **205** and the first TCO layer of the electrochromic lite is a primary seal material, **215**. This primary seal material is also between spacer **205** and second glass lite **210**. Around the perimeter of spacer **205** is a secondary seal, **220**. Bus bar wiring/leads traverse the seals for connection to a controller. Secondary seal **220** may be much thicker than depicted. These seals aid in keeping moisture out of an interior space, **225**, of the IGU. They also serve to prevent argon or other gas in the interior of the IGU from escaping.

Introduction and Overview of Electrochromic Devices: Function

FIG. 3A schematically depicts an electrochromic device, **300**, in cross-section. Electrochromic device **300** includes a substrate, **302**, a first conductive layer (CL), **304**, an electrochromic layer (EC), **306**, an ion conducting layer (IC), **308**, a counter electrode layer (CE), **310**, and a second conductive layer (CL), **314**. Layers **304**, **306**, **308**, **310**, and **314** are collectively referred to as an electrochromic stack **320**. A voltage source **316** operable to apply an electric potential across electrochromic stack **320** effects the transition of the electrochromic device from, for example, a bleached state to a colored state (depicted). The order of layers can be reversed with respect to the substrate.

Electrochromic devices having distinct layers as described can be fabricated as all solid state and/or all inorganic devices with low defectivity. Such devices and methods of fabricating them are described in more detail in U.S. patent application Ser. No. 12/645,111, entitled, "Fabrication of Low-Defectivity Electrochromic Devices," filed on Dec. 22, 2009 and naming Mark Kozlowski et al. as inventors, and in U.S. patent application Ser. No. 12/645,159, entitled, "Electrochromic Devices," filed on Dec. 22,

2009 and naming Zhongchun Wang et al. as inventors, both of which are incorporated by reference herein for all purposes. It should be understood, however, that any one or more of the layers in the stack may contain some amount of organic material. The same can be said for liquids that may be present in one or more layers in small amounts. It should also be understood that solid state material may be deposited or otherwise formed by processes employing liquid components such as certain processes employing sol-gels or chemical vapor deposition.

Additionally, it should be understood that the reference to a transition between a bleached state and colored state is non-limiting and suggests only one example, among many, of an electrochromic transition that may be implemented. Unless otherwise specified herein (including the foregoing discussion), whenever reference is made to a bleached-colored transition, the corresponding device or process encompasses other optical state transitions such as non-reflective-reflective, transparent-opaque, etc. Further, the term “bleached” refers to an optically neutral state, for example, uncolored, transparent, or translucent. Still further, unless specified otherwise herein, the “color” of an electrochromic transition is not limited to any particular wavelength or range of wavelengths. As understood by those of skill in the art, the choice of appropriate electrochromic and counter electrode materials governs the relevant optical transition.

In embodiments described herein, the electrochromic device reversibly cycles between a bleached state and a colored state. In some cases, when the device is in a bleached state, a potential is applied to the electrochromic stack **320** such that available ions in the stack reside primarily in the counter electrode **310**. When the potential on the electrochromic stack is reversed, the ions are transported across the ion conducting layer **308** to the electrochromic material **306** and cause the material to transition to the colored state.

Referring again to FIG. 3A, voltage source **316** may be configured to operate in conjunction with radiant and other environmental sensors. As described herein, voltage source **316** interfaces with a device controller (not shown in this figure). Additionally, voltage source **316** may interface with an energy management system that controls the electrochromic device according to various criteria such as the time of year, time of day, and measured environmental conditions. Such an energy management system, in conjunction with large area electrochromic devices (e.g., an electrochromic window), can dramatically lower the energy consumption of a building.

Any material having suitable optical, electrical, thermal, and mechanical properties may be used as substrate **302**. Such substrates include, for example, glass, plastic, and mirror materials. Suitable glasses include either clear or tinted soda lime glass, including soda lime float glass. The glass may be tempered or untempered.

In many cases, the substrate is a glass pane sized for residential window applications. The size of such glass pane can vary widely depending on the specific needs of the residence. In other cases, the substrate is architectural glass. Architectural glass is typically used in commercial buildings, but may also be used in residential buildings, and typically, though not necessarily, separates an indoor environment from an outdoor environment. In certain embodiments, architectural glass is at least 20 inches by 20 inches, and can be much larger, for example, as large as about 80 inches by 120 inches. Architectural glass is typically at least about 2 mm thick, typically between about 3 mm and about

6 mm thick. Of course, electrochromic devices are scalable to substrates smaller or larger than architectural glass. Further, the electrochromic device may be provided on a mirror of any size and shape.

On top of substrate **302** is conductive layer **304**. In certain embodiments, one or both of the conductive layers **304** and **314** is inorganic and/or solid. Conductive layers **304** and **314** may be made from a number of different materials, including conductive oxides, thin metallic coatings, conductive metal nitrides, and composite conductors. Typically, conductive layers **304** and **314** are transparent at least in the range of wavelengths where electrochromism is exhibited by the electrochromic layer. Transparent conductive oxides include metal oxides and metal oxides doped with one or more metals. Examples of such metal oxides and doped metal oxides include indium oxide, indium tin oxide, doped indium oxide, tin oxide, doped tin oxide, zinc oxide, aluminum zinc oxide, doped zinc oxide, ruthenium oxide, doped ruthenium oxide and the like. Since oxides are often used for these layers, they are sometimes referred to as “transparent conductive oxide” (TCO) layers. Thin metallic coatings that are substantially transparent may also be used.

The function of the conductive layers is to spread an electric potential provided by voltage source **316** over surfaces of the electrochromic stack **320** to interior regions of the stack, with relatively little ohmic potential drop. The electric potential is transferred to the conductive layers through electrical connections to the conductive layers. In some embodiments, bus bars, one in contact with conductive layer **304** and one in contact with conductive layer **314**, provide the electric connection between the voltage source **316** and the conductive layers **304** and **314**. The conductive layers **304** and **314** may also be connected to the voltage source **316** with other conventional means.

Overlaying conductive layer **304** is electrochromic layer **306**. In some embodiments, electrochromic layer **306** is inorganic and/or solid. The electrochromic layer may contain any one or more of a number of different electrochromic materials, including metal oxides. Such metal oxides include tungsten oxide (WO_3), molybdenum oxide (MoO_3), niobium oxide (Nb_2O_5), titanium oxide (TiO_2), copper oxide (CuO), iridium oxide (Ir_2O_3), chromium oxide (Cr_2O_3), manganese oxide (Mn_2O_3), vanadium oxide (V_2O_5), nickel oxide (Ni_2O_3), cobalt oxide (Co_2O_3) and the like. During operation, electrochromic layer **306** transfers ions to and receives ions from counter electrode layer **310** to cause optical transitions.

Generally, the colorization (or change in any optical property—e.g., absorbance, reflectance, and transmittance) of the electrochromic material is caused by reversible ion insertion into the material (e.g., intercalation) and a corresponding injection of a charge balancing electron. Typically some fraction of the ions responsible for the optical transition is irreversibly bound up in the electrochromic material. Some or all of the irreversibly bound ions are used to compensate “blind charge” in the material. In most electrochromic materials, suitable ions include lithium ions (Li^+) and hydrogen ions (H^+) (that is, protons). In some cases, however, other ions will be suitable. In various embodiments, lithium ions are used to produce the electrochromic phenomena. Intercalation of lithium ions into tungsten oxide (WO_{3-y} , ($0 < y \leq 0.3$)) causes the tungsten oxide to change from transparent (bleached state) to blue (colored state).

Referring again to FIG. 3A, in electrochromic stack **320**, ion conducting layer **308** is sandwiched between electrochromic layer **306** and counter electrode layer **310**. In some embodiments, counter electrode layer **310** is inorganic and/

or solid. The counter electrode layer may comprise one or more of a number of different materials that serve as a reservoir of ions when the electrochromic device is in the bleached state. During an electrochromic transition initiated by, for example, application of an appropriate electric potential, the counter electrode layer transfers some or all of the ions it holds to the electrochromic layer, changing the electrochromic layer to the colored state. Concurrently, in the case of NiWO, the counter electrode layer colors with the loss of ions.

In some embodiments, suitable materials for the counter electrode complementary to WO₃ include nickel oxide (NiO), nickel tungsten oxide (NiWO), nickel vanadium oxide, nickel chromium oxide, nickel aluminum oxide, nickel manganese oxide, nickel magnesium oxide, chromium oxide (Cr₂O₃), manganese oxide (MnO₂), and Prussian blue.

When charge is removed from a counter electrode 310 made of nickel tungsten oxide (that is, ions are transported from counter electrode 310 to electrochromic layer 306), the counter electrode layer will transition from a transparent state to a colored state.

In the depicted electrochromic device, between electrochromic layer 306 and counter electrode layer 310, there is the ion conducting layer 308. Ion conducting layer 308 serves as a medium through which ions are transported (in the manner of an electrolyte) when the electrochromic device transitions between the bleached state and the colored state. Preferably, ion conducting layer 308 is highly conductive to the relevant ions for the electrochromic and the counter electrode layers, but has sufficiently low electron conductivity that negligible electron transfer takes place during normal operation. A thin ion conducting layer with high ionic conductivity permits fast ion conduction and hence fast switching for high performance electrochromic devices. In certain embodiments, the ion conducting layer 308 is inorganic and/or solid.

Examples of suitable ion conducting layers (for electrochromic devices having a distinct IC layer) include silicates, silicon oxides, tungsten oxides, tantalum oxides, niobium oxides, and borates. These materials may be doped with different dopants, including lithium. Lithium doped silicon oxides include lithium silicon-aluminum-oxide. In some embodiments, the ion conducting layer comprises a silicate-based structure. In some embodiments, a silicon-aluminum-oxide (SiAlO) is used for the ion conducting layer 308.

Electrochromic device 300 may include one or more additional layers (not shown), such as one or more passive layers. Passive layers used to improve certain optical properties may be included in electrochromic device 300. Passive layers for providing moisture or scratch resistance may also be included in electrochromic device 300. For example, the conductive layers may be treated with anti-reflective or protective oxide or nitride layers. Other passive layers may serve to hermetically seal electrochromic device 300.

FIG. 3B is a schematic cross-section of an electrochromic device in a bleached state (or transitioning to a bleached state). In accordance with specific embodiments, an electrochromic device, 400, includes a tungsten oxide electrochromic layer (EC), 406, and a nickel-tungsten oxide counter electrode layer (CE), 410. Electrochromic device 400 also includes a substrate, 402, a conductive layer (CL), 404, an ion conducting layer (IC), 408, and conductive layer (CL), 414.

A power source, 416, is configured to apply a potential and/or current to an electrochromic stack, 420, through suitable connections (e.g., bus bars) to the conductive layers,

404 and 414. In some embodiments, the voltage source is configured to apply a potential of about 2 volts in order to drive a transition of the device from one optical state to another. The polarity of the potential as shown in FIG. 3A is such that the ions (lithium ions in this example) primarily reside (as indicated by the dashed arrow) in nickel-tungsten oxide counter electrode layer 410.

FIG. 3C is a schematic cross-section of electrochromic device 400 shown in FIG. 3B but in a colored state (or transitioning to a colored state). In FIG. 3C, the polarity of voltage source 416 is reversed, so that the electrochromic layer is made more negative to accept additional lithium ions, and thereby transition to the colored state. As indicated by the dashed arrow, lithium ions are transported across ion conducting layer 408 to tungsten oxide electrochromic layer 406. Tungsten oxide electrochromic layer 406 is shown in the colored state. Nickel-tungsten oxide counter electrode 410 is also shown in the colored state. As explained, nickel-tungsten oxide becomes progressively more opaque as it gives up (deintercalates) lithium ions. In this example, there is a synergistic effect where the transition to colored states for both layers 406 and 410 are additive toward reducing the amount of light transmitted through the stack and substrate.

As described above, an electrochromic device may include an electrochromic (EC) electrode layer and a counter electrode (CE) layer separated by an ionically conductive (IC) layer that is highly conductive to ions and highly resistive to electrons. As conventionally understood, the ionically conductive layer therefore prevents shorting between the electrochromic layer and the counter electrode layer. The ionically conductive layer allows the electrochromic and counter electrodes to hold a charge and thereby maintain their bleached or colored states. In electrochromic devices having distinct layers, the components form a stack which includes the ion conducting layer sandwiched between the electrochromic electrode layer and the counter electrode layer. The boundaries between these three stack components are defined by abrupt changes in composition and/or microstructure. Thus, the devices have three distinct layers with two abrupt interfaces.

In accordance with certain embodiments, the counter electrode and electrochromic electrodes are formed immediately adjacent one another, sometimes in direct contact, without separately depositing an ionically conducting layer. In some embodiments, electrochromic devices having an interfacial region rather than a distinct IC layer are employed with angled bus bars described herein. Such devices, and methods of fabricating them, are described in U.S. patent application Ser. Nos. 12/772,055 and 12/772,075, each filed on Apr. 30, 2010, and in U.S. patent application Ser. Nos. 12/814,277 and 12/814,279, each filed on Jun. 11, 2010—each of the four applications is entitled “Electrochromic Devices,” each names Zhongchun Wang et al. as inventors, and each is incorporated by reference herein in its entirety.

FIG. 4 is a schematic cross-section of an electrochromic device, 450, in a colored state, where the device has an interfacial region, 418, which does not contain a distinct IC layer. Voltage source 416, conductive layers 414 and 404, and substrate 402 are essentially the same as described in relation to FIGS. 3A-3C. Between conductive layers 414 and 404 is a region 420, which includes counter electrode layer 410, electrochromic layer 406, and interfacial region 418 between them, rather than a distinct IC layer. In this example, there is no distinct boundary between counter electrode layer 410 and interfacial region 418, nor is there a

distinct boundary between electrochromic layer **406** and interfacial region **418**. Rather, there is a diffuse transition between CE layer **410** and interfacial region **418**, and between interfacial region **418** and EC layer **406**.

Transition of an Electrochromic Device

As noted above, a switchable optical device such as an electrochromic device reversibly cycles between two or more optical states such as a bleached state and a colored state. Switching between these states is controlled by applying predefined current and/or voltage to the device. Higher currents and/or voltages are generally applied to larger switchable optical devices compared to smaller switchable optical devices to cycle between optical states.

The speed of coloration of an electrochromic device is a function of the applied current and/or voltage. Generally, the higher the current and/or voltage applied across the electrochromic device, the faster the electrochromic device will transition between optical states. Applying a high current and/or voltage across the electrochromic device, however, depending on the configuration of the bus bars of the electrochromic device, may result in damage to regions of the electrochromic device due to the regions being exposed to too high a current and/or voltage. These regions are sometimes referred to as "hot spots." On the other hand, if a low current and/or voltage is applied across the electrochromic device, the device may not completely switch between optical states or may switch at an undesirably slow rate between optical states.

Further, large electrochromic devices such as those on residential windows or architectural glass may exhibit an effect sometimes referred to as the "terminal effect." This is due to the relatively high sheet resistance of thin electrochromic device films, including the electrodes, coupled with device designs having terminals (bus bars) located outside the viewable area of the substrate, for example, only at the edges of the device/substrate. In such devices, there is a considerable ohmic potential drop over the area of the device (and concomitant leakage current), from the terminal located at the edge of the device to the center of the device, where there is no contact to an external voltage source. As a result, not only does the center region of the device transition more slowly than the edge regions proximate terminal contacts (which is referred to as the "curtain effect"), but the center region may never fully transition. In other words, the center may transition only to a limited extent in comparison to the edges. Further, to the extent that the center of the device transitions, it does so more slowly than the edge of the device. Yet further, it may be difficult to hold a transition state, once such state is reached in the center of the device. These edge-to-center non-uniformities can be perceptible to and distracting to users.

Bus Bar Configurations

Different bus bar configurations may be used, e.g., with a large electrochromic device, to substantially minimize hot spots or exposing the electrochromic device to a damaging current and/or voltage and to substantially minimize the curtain effect. Additionally, different bus bar configurations may be used to substantially maximize the transition rate between optical states of the electrochromic device.

FIG. 5A shows a top-down view of an electrochromic lite, **500**, including bus bars having a planar configuration. In some embodiments, electrochromic lite **500** may be similar to electrochromic lite **100** shown in FIGS. 1A-1C. Electrochromic lite **500** includes a first bus bar, **505**, disposed on a first conductive layer, **510**, and a second bus bar, **515**, disposed on a second conductive layer, **520**. An electrochromic stack (not shown) is between first conductive layer **510**

and second conductive layer **520**. As shown, first bus bar **505** may extend substantially across one side of first conductive layer **510**. Second bus bar **515** may extend substantially across one side of second conductive layer **520** opposite the side of electrochromic lite **500** on which first bus bar **505** is disposed.

As noted above, the configuration of first and second bus bars **505** and **515** in FIG. 5A may be referred to as a planar bus bar configuration. FIG. 5B is a graph showing a plot of the voltage applied to first bus bar **505** and the voltage applied to second bus bar **515** that may be used to transition electrochromic lite **500** from a bleached state to a colored state, for example. Plot **525** shows the voltage applied to first bus bar **505** and through first conductive layer **510**. As shown, the voltage drops from the left hand side (e.g., where first bus bar **505** is disposed on first conductive layer **510** and where the voltage is applied) to the right hand side of first conductive layer **510** due to the sheet resistance of first conductive layer **510**. Plot **530** shows the voltage applied to second bus bar **515** and through second conductive layer **520**. As shown, the voltage increases from the right hand side (e.g., where second bus bar **515** is disposed on second conductive layer **520** and where the voltage is applied) to the left hand side of second conductive layer **520** due to the sheet resistance of second conductive layer **520**.

FIG. 5C is a graph showing a plot of the effective voltage applied across the electrochromic device between first and second conductive layers **510** and **520** of electrochromic lite **500**. The effective voltage is the voltage difference between the voltage applied to first bus bar **505** and through first conductive layer **510** and the voltage applied to second bus bar **515** and through second conductive layer **520**. Regions of an electrochromic device subjected to higher effective voltages transition between optical states faster than regions subjected to lower effective voltages. As shown, the effective voltage is the lowest at the center of electrochromic lite **500** and highest at the edges of electrochromic lite **500**.

FIG. 5D is a graph showing plots of the percent absorption of visible light through electrochromic lite **500** when it is transitioning from a bleached state to a colored state due to the application of the voltages applied to first and second bus bars **505** and **515** in FIG. 5B. Plot **540** is the percent absorption across electrochromic lite **500** a short time after application of the voltages. In plot **540** the edges of electrochromic lite **500** have transitioned faster than the center of electrochromic lite **500**. This is the curtain effect; coloration of electrochromic lite **500** starts along the edges of electrochromic lite **500** where first and second bus bars **505** and **515** are disposed. Plot **545** shows the percent absorption across electrochromic lite **500**, midway through the transition from a bleached state to a colored state. In plot **545**, electrochromic lite **500** is more colored (i.e., higher percent absorption of light), but there is still a drop in the absorption of light at the center of electrochromic lite **500**. Plot **550** is the percent absorption across electrochromic lite **500** when the lite is in a fully colored state. As shown, the percent absorption of light may not be uniform across electrochromic lite **500** in the fully colored state due to the effective voltage drop across electrochromic lite **500** (see FIG. 5C), but this may not be perceptible to the human eye.

An advantage of planar bus bar configurations includes little risk of overdriving an electrochromic device. However, planar bus bar configurations may exhibit a longer duration curtain effect and take longer to transition an electrochromic device between optical states than other bus bar configurations.

13

FIG. 6A shows a top-down view of an electrochromic lite, **600**, including bus bars having a cylindrical configuration. Electrochromic lite **600** includes first bus bars, **605** and **607**, disposed on a first conductive layer, **610**, and second bus bars, **615** and **617**, disposed on a second conductive layer, **620**. An electrochromic stack (not shown) is between first conductive layer **610** and second conductive layer **620**. First bus bars **605** and **607** extend substantially across two opposing sides of first conductive layer **610**. Second bus bars **615** and **617** extend substantially across two opposing sides of second conductive layer **620**. As shown, first bus bars **605** and **607** and second bus bars **615** and **617** may be at locations such that first bus bar **605** and second bus bar **615** are close to one another and first bus bar **607** and second bus bar **617** are close to one another.

As noted above, the configuration of first bus bars **605** and **607** and second bus bars **615** and **617** may be referred to as a cylindrical bus bar configuration. FIG. 6B is a graph showing a plot of the voltage applied to first bus bars **605** and **607** and the voltage applied to second bus bars **615** and **617** that may be used to transition electrochromic lite **600** from a bleached state to a colored state, for example. The same voltage may be applied to both of the first bus bars **605** and **607** (i.e., first bus bars **605** and **607** may be connected in parallel). The same voltage may be applied to both of the second bus bars **615** and **617** (i.e., second bus bars **615** and **617** may be connected in parallel). Plot **625** shows the voltage applied to first bus bars **605** and **607** and through first conductive layer **610**. As shown, the voltage drops from the left hand side (e.g., where first bus bar **605** is disposed on first conductive layer **610** and where the voltage is applied) to the center of first conductive layer **610** due to the sheet resistance of first conductive layer **610**. The voltage increases from the center of first conductive layer **610** to the right hand side (e.g., where first bus bar **607** is disposed on first conductive layer **610** and where the voltage is applied).

Plot **630** shows the voltage applied to second bus bars **615** and **617** and through second conductive layer **620**. As shown, the voltage increases from the left hand side (e.g., where second bus bar **615** is disposed on second conductive layer **620** and where the voltage is applied) to the center of second conductive layer **620** due to the sheet resistance of second conductive layer **620**. The voltage decreases from the center of second conductive layer **620** to the right hand side (e.g., where second bus bar **617** is disposed on second conductive layer **620** and where the voltage is applied).

FIG. 6C is a graph showing a plot of the effective voltage applied across the electrochromic device between first and second conductive layers **610** and **620** of electrochromic lite **600**. As shown, the effective voltage is the lowest at the center of electrochromic lite **600** and highest at the edges of electrochromic lite **600**; the effective voltage at the edges of electrochromic lite **600** with the cylindrical bus bars is higher than the effective voltage at the edges of electrochromic lite **500** due to bus bars **605** and **615** and bus bars **607** and **617** being close to one another. While the edges of electrochromic lite **600** may transition between optical states rapidly due to this high effective voltage, the high effective voltage may overdrive the electrochromic device at the edges. Overdriving the device may mean that the device is damaged due to the inability of the device's structure to handle excess voltage, current, and/or ion flow. For example, the device may delaminate and/or cease to function in areas where overdriven.

FIG. 6D is a graph showing plots of the percent absorption of visible light through electrochromic lite **600** when it is transitioning from a bleached state to a colored state due

14

to the application of the voltages applied to first bars **605** and **607** and second bus bars **615** and **617** in FIG. 6B. Plot **640** is the percent absorption across electrochromic lite **600** a short time after application of the voltages. In plot **640** the edges of electrochromic lite **600** have transitioned faster than the center of electrochromic lite **600**; the edges would transition faster than the edges of electrochromic lite **500**. Plot **645** is the percent absorption across electrochromic lite **600**, midway through the transition from a bleached state to a colored state. In plot **645**, electrochromic lite **600** is more colored (i.e., higher percent absorption of light), but there is still a drop in the absorption of light at the center of electrochromic lite **600**. Plot **650** is the percent absorption across electrochromic lite **600** when the lite is in a fully colored state. As shown, the percent absorption of light may not be uniform across electrochromic lite **600** in the fully colored state due to the effective voltage drop across electrochromic lite **600** (see FIG. 6C), but this may not be perceptible to the human eye.

Advantages of cylindrical bus bar configurations include a shorter duration curtain effect and a shorter transition time of an electrochromic compared to planar bus bar configurations. The first and second bus bars being close to one another, however, may cause hot spots in the electrochromic device.

FIG. 7A shows a top-down view of an electrochromic lite, **700**, including bus bars having a perpendicular cylindrical configuration. Electrochromic lite **700** includes first bus bars, **705** and **707**, disposed on a first conductive layer, **710**, and second bus bars, **715** and **717**, disposed on a second conductive layer, **720**. An electrochromic stack (not shown) is between first conductive layer **710** and second conductive layer **720**. First bus bars **705** and **707** extend substantially across two opposing sides of first conductive layer **710**. Second bus bars **715** and **717** extend substantially across two opposing sides of second conductive layer **720**. As shown, first bus bars **705** and **707** and second bus bars **715** and **717** are at locations such that the first bus bars are substantially perpendicular to the second bus bars. To transition electrochromic lite **700** between optical states, the same voltage may be applied to both of first bus bars **705** and **707** (i.e., first bus bars **705** and **707** may be connected in parallel). The same voltage may be applied to both of second bus bars **715** and **717** (i.e., second bus bars **715** and **717** may be connected in parallel).

FIG. 7B shows the coloration of electrochromic lite **700** at a time midway through the transition from a bleached state to a colored state. As shown in FIG. 7B, the corners of electrochromic lite **700** color first due to high effective voltages at the corners. For example, the high effective voltage at a corner, **730**, of electrochromic lite **700** is due to first bus bar **707** and second bus bar **715** being close to one another at corner **730**. The center of electrochromic lite **700** is the last portion of the electrochromic lite to transition from the bleached state to the color state.

An advantage of perpendicular cylindrical bus bar configurations is that they may allow for more symmetric coloration (i.e., from the corners to the center) of an electrochromic lite. However, perpendicular cylindrical bus bar configurations are prone to hot spots at the corners of an electrochromic lite and damage the electrochromic device, because of the proximity of oppositely polarized bus bars at the corners of the device. For example, to obtain the desired coloration in the center of an electrochromic lite, the corners may be overdriven, which increases the leakage current of the electrochromic device as well as the likelihood of device damage/degradation.

Angled Bus Bars

Instead of the planar, cylindrical, or perpendicular cylindrical bus bar configurations described above with respect to FIG. 5A-7B, an angled bus bar may be included in some embodiments of electrochromic lites or other optically switchable devices. An angled bus bar may include two arms that meet at an end of each of the arms, forming an angle between the two arms. For example, the two arms may form an L-shape or a V-shape. In some embodiments, an electrochromic lite including at least one angled bus bar may transition between optical states faster and exhibit a more symmetric transition between optical states. Further, an angled bus bar may be configured such that hot spots in an electrochromic device are substantially minimized.

FIG. 8A shows a top-down view of a square-shaped electrochromic lite, **800**, including angled bus bars. Electrochromic lite **800** is depicted in the bleached state in FIG. 8A. Electrochromic lite **800** includes a first bus bar, **805**, disposed on a first conductive layer, **810**, and a second bus bar, **815**, disposed on a second conductive layer, **820**. An electrochromic stack (not shown) is between first conductive layer **810** and second conductive layer **820**. As shown, first conductive layer **810** and second conductive layer **820** both have a perimeter with corners including sides and vertices. The electrochromic stack between first conductive layer **810** and second conductive layer **820** may also have a perimeter with corners including sides and vertices similar to first conductive layer **810** and second conductive layer **820**.

First bus bar **805** is disposed on first conductive layer **810** proximate to a corner of first conductive layer **810**. First bus bar **805** include a first arm, **806**, and a second arm, **807**, that substantially follow the shape of a first side, a first vertex, and a second side of a corner of first conductive layer **810**, respectively. Second bus bar **815** is disposed on second conductive layer **820** proximate to a corner of second conductive layer **820**. Second bus bar **815** includes a first arm, **816**, and a second arm, **817** that substantially follow the shape of a first side, a first vertex, and a second side of a corner of second conductive layer **820**, respectively. In some embodiments, a first bus bar and a second bus bar may have configurations that are substantially similar. For example, in some embodiments, the arms of the first and second bus bars are of substantially equal length. In some embodiments, a first bus bar and a second bus bar may be substantially symmetric to one another.

In some embodiments, the first and the second arms of the bus bars may span about 10% to 90%, about 10% to 65%, or about 35% to 65% of the length of the respective sides of the electrochromic device area. In some embodiments, a first arm of a first bus bar has a length about 10% to 40%, about 35% to 65%, or about 60% to 90% of a length of the first side of the electrochromic device. In some embodiments, the first and the second arms of the bus bars may be substantially perpendicular. In some embodiments, each of the first and second bus bars may span sides defining diagonally opposing corners of the electrochromic lite.

FIG. 8B shows the coloration of electrochromic lite **800** at a time midway through the transition from a bleached state to a colored state. As shown in FIG. 8B, corners, **830** and **835**, of electrochromic lite **800** where bus bars **805** and **815** are disposed, color first. The center band of electrochromic lite **800**, from corners, **840** and **845**, is the last portion of electrochromic lite **800** to transition from the bleached state to the colored state.

While first conductive layer **810** and second conductive layer **820** are shown as having a substantially square shape in electrochromic lite **800**, the conductive layers and the

electrochromic stack disposed there between may have any shape, including other polygons or rectangles.

FIG. 8C shows a top-down view of a rectangular electrochromic lite, **850**, including angled bus bars. Electrochromic lite **850** includes a first bus bar, **855**, disposed on a first conductive layer, **860**, and a second bus bar, **865**, disposed on a second conductive layer, **870**. An electrochromic stack (not shown) is between first conductive layer **860** and second conductive layer **870**. First bus bar **855** is disposed on first conductive layer **860** proximate to a corner of first conductive layer **860**. First bus bar **855** includes a first arm, **856**, and a second arm, **857**, which substantially follow the shape of a first side, a first vertex, and a second side of a corner of first conductive layer **860**. Second bus bar **865** is disposed on second conductive layer **870** proximate to a corner of second conductive layer **870**. Second bus bar **865** includes a first arm, **866**, and a second arm, **867**, which substantially follow the shape of a first side, a first vertex, and a second side of a corner of second conductive layer **870**. First and second bus bars **855** and **865** are disposed on diagonally opposing corners of electrochromic lite **850**. In some embodiments, a first bus bar and a second bus bar may have configurations that are substantially similar.

In some embodiments, second arm **857** and **867** of each of first and second bus bars **855** and **865**, respectively, traverse a portion of a longer side of electrochromic lite **850**. First arms **856** and **866** of each of first and second bus bars **855** and **865**, respectively, traverse a portion of a shorter side of electrochromic lite **850**. In some embodiments, the first arms and the second arms of first and second bus bars **855** and **865** are substantially orthogonal to one another. In some embodiments, first arms **856** and **866** span about 25% to 75% of the shorter side. In some embodiments, second arms **857** and **867** span about 75% to 90% of the longer side.

In some embodiments, including at least one angled bus bar in an electrochromic lite, as in electrochromic lites **800** and **850** described above with respect to FIGS. 8A and 8C, may allow for semi-symmetric coloration (i.e., from two corners to the center) of an electrochromic lite. When transitioning from a bleached state to a colored state, all sides of the electrochromic lite may begin to color at similar rates (i.e., side preference coloration may be substantially eliminated), which reduces the asymmetry of the curtain effect. Further, the hot spots generated by, for example, perpendicular cylindrical bus bar configurations are avoided in angled bus bar configurations. Further, the transition between optical states may occur more rapidly with angled bus bar configurations than with planar bus bar configurations without overdriving regions of the electrochromic lite, as with some cylindrical or perpendicular cylindrical bus bar configurations. FIGS. 9-12B show top-down views of electrochromic lites including angled bus bars in different configurations.

Referring to FIG. 9, an electrochromic lite, **900**, includes first bus bars, **930** and **935**, disposed on a first conductive layer, **910**, and second bus bars, **940** and **945**, disposed on a second conductive layer, **920**. An electrochromic stack (not shown) is between first conductive layer **910** and second conductive layer **920**. As shown, the first conductive layer **910** and the second conductive layer **920** both have a perimeter with corners including sides and vertices. The electrochromic stack between first conductive layer **910** and second conductive layer **920** may also have a perimeter with corners including sides and vertices similar to first conductive layer **910** and second conductive layer **920**.

First bus bar **930** is disposed on first conductive layer **910** proximate to a corner of first conductive layer **910**. First bus

bar **935** is disposed on first conductive layer **910** proximate to a corner of first conductive layer **910** diagonally opposed to first bus bar **930**. Similarly, second bus bar **940** is disposed on second conductive layer **920** proximate to a corner of second conductive layer **920**. Second bus bar **945** is disposed on second conductive layer **920** proximate to a corner of second conductive layer **920** diagonally opposed to second bus bar **940**. To transition electrochromic lite **900** between optical states, the same voltage may be applied to both of first bus bars **930** and **935** (i.e., first bus bars **930** and **935** may be connected in parallel). The same voltage may be applied to both of second bus bars **940** and **945** (i.e., second bus bars **940** and **945** may be connected in parallel).

Each of the bus bars includes arms that follow the shapes of the sides and the vertex of the corner with which the bus bar is associated. Arms of first bus bars **930** and **935** are not in close proximity to arms of second bus bars **940** and **945** in order to avoid hot spots, in some embodiments. For example, arm, **941**, of second bus bar **940** extends along the side of second conductive layer **920**. Arm, **936**, of first bus bar **935** extends along the side of first conductive layer **910**, with these sides of first conductive layer **910** and second conductive layer **920** being substantially parallel. In some embodiments, the ends of arms **941** and **936** do not overlap with one another (e.g., as the bus bars in a cylindrical configuration do), and in some other embodiments, the ends of arms **941** and **936** are not in close proximity to one another. The length of arm, **942**, of second bus bar **940** and the length of arm, **932**, of first bus bar **930**, may be specified in a similar manner in order to avoid hot spots.

FIG. **10** shows a top-down view of an electrochromic lite, **1000**, including angled bus bars. The bus bar configuration of electrochromic lite **1000** is similar to the bar configuration of electrochromic lite **900** shown in FIG. **9**, but instead of the first bus bars being diagonally opposed to one another, each of the first bus bars is diagonally opposed to a second bus bar. Electrochromic lite **1000** includes first bus bars, **1030** and **1035**, disposed on a first conductive layer, **1010**, and second bus bars, **1040** and **1045**, disposed on a second conductive layer, **1020**. An electrochromic stack (not shown) is between first conductive layer **1010** and second conductive layer **1020**. As shown, the first conductive layer **1010** and the second conductive layer **1020** both have a perimeter with corners including sides and vertices. The electrochromic stack between first conductive layer **1010** and second conductive layer **1020** may also have a perimeter with corners including sides and vertices similar to first conductive layer **1010** and second conductive layer **1020**.

First bus bar **1030** is disposed on first conductive layer **1010** proximate to a corner of first conductive layer **1010**. Second bus bar **1040** is disposed on second conductive layer **1020** proximate to a corner of second conductive layer **1020** diagonally opposed to first bus bar **1030**. Similarly, first bus bar **1035** is disposed on first conductive layer **1010** proximate to a corner of first conductive layer **1010**. Second bus bar **1045** is disposed on second conductive layer **1020** proximate to a corner of second conductive layer **1020** diagonally opposed to first bus bar **1035**. To transition electrochromic lite **1000** between optical states, the same voltage may be applied to both of first bus bars **1030** and **1035** (i.e., first bus bars **1030** and **1035** may be connected in parallel). The same voltage may be applied to both of second bus bars **1040** and **1045** (i.e., second bus bars **1040** and **1045** may be connected in parallel).

Again, similar to electrochromic lite **900**, each of the bus bars of electrochromic lite **1000** includes arms that follow the shapes of the sides and the vertex of the corner with

which the bus bar is associated. The end portion of the arms of first bus bars **1030** and **1035** are not in close proximity to the ends of the arms of second bus bars **1040** and **1045** in order to avoid hot spots, in some embodiments. For example, arm, **1032**, of first bus bar **1030** extends along the side of first conductive layer **1010**. Arm, **1047**, of second bus bar **1045** extends along the side of second conductive layer **1020**. In some embodiments, the ends of arms **1032** and **1047** do not overlap with one another (e.g., as the bus bars in a cylindrical configuration do), and in some other embodiments, the ends of arms **1032** and **1047** are not in close proximity to one another. The lengths of the arms of first bus bar **1035** and second bus bar **1040** may be specified in a similar manner to avoid hot spots. In another embodiment, the bus bars depicted in FIG. **10** are arranged such that bus bar **1040** and **1045**, while still on the lower electrode, are diagonally opposed, and bus bars **1030** and **1035**, while remaining on the upper electrode, are also diagonally opposed.

Arm, **1031**, of first bus bar **1030** and arm, **1036**, of first bus bar **1035**, along the same side of first conductive layer **1010**, may have lengths such that the electrochromic lite **1000** transitions between optical states in a short time period of time. The arms of second bus bars **1040** and **1045** along the same side of second conductive layer **1020** also may have lengths such that the electrochromic device transitions between optical states in a short time period of time. In some embodiments, the arms of the first bus bars may meet, forming a single C-shaped first bus bar. In some embodiments, the arms of the second bus bars may meet, forming a single C-shaped second bus bar. This embodiment is shown in FIG. **11**. Electrochromic lite **1100** shown in FIG. **11** includes a first bus bar, **1130**, disposed on a first conductive layer, **1110**, and a second bus bar, **1140**, disposed on a second conductive layer, **1120**. Note, for example, as described above in relation to FIG. **11**, an angled bus bar may have more than one angled feature (see bus bars **1130** and **1140**, each having two angles). One embodiment is an angled bus bar having two or more angles.

The angle between the two arms of an angled bus bar may approach 180 degrees and may also be smaller than 90 degrees. For the purposes of illustration only, the embodiments above are described in terms of rectangular electrochromic lites, however, non-rectangular shapes and other optical devices are meant to be included within the scope of the disclosed embodiments. FIG. **12A** depicts top view schematics of a number of electrochromic lites of varying shapes and angled bus bar configurations. For example, an electrochromic window, **1200**, has a trapezoidal shape and two bus bars. A lower transparent electrode surface, **1205**, bears an angled bus bar, **1215**, while an upper transparent electrode, **1210**, has another angled bus bar, **1220**. In this example, bus bar **1215** has an angle larger than 90 degrees, while bus bar **1220** has an acute angle, less than 90 degrees. Depending upon the shape of the electrochromic lite, the bus bars may or may not be symmetrical or span the same length on each arm or have the same total length. As well, an electrochromic lite may have a combination of angled and non-angled bus bars. For example, triangular lite, **1225**, has an angled bus bar, **1240**, on a lower electrode **1230**, and a linear bus bar, **1245**, on an upper electrode, **1235**. In another example, triangular electrochromic lite, **1250**, has a single angled bus bar, **1265**, on a lower electrode, **1255**, while also having two angled bus bars, **1270**, on an upper electrode, **1260**. The shape and orientation of the bus bars will depend on the shape of the window, the switching and/or energy requirements of the device they power, and the like, e.g., in

order to maximize efficient switching and/or powering, homogeneity in transitions, and to minimize hotspots. In another example, a hexagonal electrochromic lite, **1275**, has a first angled bus bar, **1286**, on a lower electrode, **1280**, and a second angled bus bar, **1287**, on an upper electrode, **1285**. In this example, angled bus bars **1286** and **1287** each have three angle features (vertices). In another example, a hexagonal electrochromic lite, **1290**, has six angled bus bars. On a lower electrode, **1292**, there are three angled bus bars, **1296**, and on an upper electrode, **1294**, there are also three angled bus bars, **1298**. Driving algorithms for electrochromic lites are described in more detail below, particularly, in relation to angled bus bars and devices having more than two bus bars.

Angled bus bars are described herein in terms of having at least a first arm and a second arm joined at a vertex, e.g., as depicted and described in relation to FIG. **12A** (some examples in FIG. **12A** illustrate that there may be more vertices and arms in certain embodiments). In certain embodiments, angled bus bars do not have a physical vertex. Referring to FIG. **12B**, there are configurations that are equivalent, and therefore considered, angled bus bars for the purposes of this disclosure. For example, electrochromic device **1201** has associated bus bar arms, **1202** and **1203**, which are electrically connected via a wire **1204**. Wire **1204** could also be a thin strip of bus bar material or other suitable electrical connector. Although there is not a physical vertex per se, this assembly is an angled bus bar because there is a virtual vertex, that is, a theoretical point in space that is an intersection of two axes, each axis passing through the length of each bus bar arm. Likewise, similarly situated bus bar arms electrically connected by a wire, **1206**, at their distal ends would also be an angled bus bar. Note, in FIG. **12B**, the electrical source can be connected via the wire or other connector, or on one (or both) of the bus bar arms. In one embodiment, if the proximal ends of two bus bar arms are: 1) at some angle relative to each other as described herein for bus bar arms having a physical vertex, 2) within about 1 mm and about 100 mm of each other, 3) in electrical communication with each other, and 4) on the same conductive layer of a device, then they are, collectively, an angled bus bar. In another embodiment, if two bus bar arms: 1) each span about 10% to 90%, about 10% to 65%, or about 35% to 65% of the length of two sides of an electrochromic device area, 2) are at some angle relative to each other as described herein for bus bar arms having a physical vertex, 3) are in electrical communication with each other, and 4) are on the same conductive layer of a device, then they are, collectively, an angled bus bar. For the purposes of these definitions, "electrical communication with each other" does not include the electrical communication via the conductive layer on which the pair of bus bar arms reside.

In certain embodiments, the bus bar arms need not be in electrical communication with each other. For example, in one embodiment, if two bus bar arms: 1) each span about 10% to 90%, about 10% to 65%, or about 35% to 65% of the length of two sides of an electrochromic device area, 2) are at some angle relative to each other as described herein for bus bar arms having a physical vertex, 3) are on the same conductive layer of a device, and 4) are independently held at the same polarity, either a negative or a positive polarity, then they are, collectively, an angled bus bar. In another embodiment, if the proximal ends of two bus bar arms are: 1) at some angle relative to each other as described herein for bus bar arms having a physical vertex, 2) within about 1 mm and about 100 mm of each other, 3) are independently held at the same polarity, either a negative or a positive

polarity, and 4) on the same conductive layer of a device, then they are, collectively, an angled bus bar. In these embodiments, is it useful, but not necessary, if the polarity applied to each of the angled bus bar's arms is similar in magnitude. These angled bus bars, although requiring more complex drive circuits, will transition the electrochromic device as those having a physical vertex or one having a virtual vertex and having the arms electrical communication with each other. The objectives of avoiding hotspots and improving optical transitions remains the same, and are met with appropriate configurations as described herein in relation to angled bus bars having physical vertices.

In some embodiments, bus bar configurations described herein may not develop hot spots in an optically switchable device while transitioning the optically switchable device between optical states. Further, the bus bar configurations described herein may be modified and/or combined, depending on the application. For example, in some embodiments, an optically switchable device may have the shape of a polygon, a trapezoid, or a shape with one region that substantially forms a corner with other regions having curved borders; bus bars may be disposed on the first and second conductive layers in a manner that substantially minimizes hot spots, substantially minimizes the curtain effect, and quickly transitions the optically switchable device between optical states.

In some embodiments, the lengths of the first bus bars along the edges of the first conductive layer and the lengths of the second bus bars along the edges of the second conductive layer are substantially maximized without developing hot spots in an optically switchable device between the first and second conductive layers. Substantially maximizing the lengths of the first and the second bus bars may aid in quickly transitioning the optically switchable device between optical states.

Generally, for fast switching between optical states of an optically switchable device, the first and the second bus bars should be as close to one another as possible, due to the inherent resistance of the conducting layers (e.g., the TCO) of the device. Having the first and the second bus bars close to each other, however, generates hot spots in the optically switchable device, as explained above. In some embodiments, the first and the second bus bars are configured such that distances between extremities of the first bus bar and portions of the second bus bar are substantially minimized to aid in quickly transitioning the optically switchable device without developing hot spots. Similarly, distances between extremities of the second bar and portions of the first bus bar are substantially minimized. For example, referring to FIG. **8A**, a corner, A, of first bus bar **805** is an extremity of first bus bar **805**. It is a point of first bus bar **805** that is furthest away from any part of second bus bar **815**. The portions of second bus bar **815** that are closest to corner A of first bus bar **805** are ends, B and C, of arms **816** and **817**, respectively, of second bus bar **815**. Thus, to aid in quickly transitioning between optical states, the distance between corner A of first bus bar **805** and either end B or C of second bus bar **815** may be substantially minimized.

Methods of Changing Optical States

FIG. **13** shows a flowchart depicting a method of changing an optical state of an optically switchable device. The method **1300** shown in FIG. **13** describes changing an optical state of an optically switchable device that may be similar to the electrochromic lite **800** shown in FIG. **8A** or the electrochromic lite **850** shown in FIG. **8C**. That is, method **1300** describes changing an optical state of an optically switchable device disposed on a surface of a

21

substrate and having a perimeter with at least one corner including a first side, a first vertex joining the first side and a second side, and the second side. Method **1300** may be used to transition any of the optically switchable devices including an angled bus bar disclosed herein.

Starting with operation **1305** of method **1300**, current and/or voltage of a first polarity is applied to a first bus bar affixed to the optically switchable device proximate to the corner. The first bus bar includes a first arm and a second arm having a configuration that substantially follows the shape of the first side, the first vertex, and the second side of the corner. For example, the first bus bar referred to in operation **1305** could be first bus bar **805** of electrochromic lite **800** shown in FIG. **8A**.

At operation **1310**, current and/or voltage of a second polarity is applied to a second bus bar affixed to the optically switchable device. The second bus bar referred to in operation **1310** could be second bus bar **815** of electrochromic lite **800** shown in FIG. **8A**. However, as described in operation **1310**, the second bus bar could have any number of different configurations, and is not necessarily an angled bus bar.

At operation **1315**, in response to the current and/or voltage applied to the first and the second bus bars, an optical state of the optically switchable device switches such that the switching occurs throughout the optically switchable device substantially uniformly. In some embodiments, having at least one angled bus bar included in an optically switchable device may allow the optically switchable device to transition between optical states more uniformly than the planar, cylindrical, or perpendicular cylindrical bus bar configurations described with respect to FIGS. **5A-7B**.

Having at least one angled bus bar included in an optically switchable device may also allow the optically switchable device to switch between optical states faster than the planar, cylindrical, or perpendicular cylindrical bus bar configurations. For example, in some embodiments, switching the optical state of the optically switchable device from a first optical state to a second optical state is performed in about 10 minutes or less. Of course, the switching may depend on a number of factors, e.g., the area of the electrochromic device, the sheet resistance of the conductor layers, etc.; however, in one embodiment, the 10 minute or less switching speed applies to an optically switchable device having an area of between about 1 square foot (0.09 m^2) and about 60 square feet (5.57 m^2), in another embodiment between about 6 square feet (0.56 m^2) and about 30 square feet (2.79 m^2), and in yet another embodiment, between about 10 square feet (0.93 m^2) and about 20 square feet (1.86 m^2). Further, having at least one angled bus bar included in an optically switchable device may allow the optically switchable device to switch between optical states without developing hot spots in the optically switchable device. For example, switching the optical state of the optically switchable device may be performed such that the current and/or the voltage applied across the optically switchable device does not damage the optically switchable device.

When more than two bus bars are configured on a device, one embodiment includes a driving algorithm that drives successive pairs of bus bars. For example, referring to FIG. **12**, electrochromic lite **1290**, a driving algorithm may include powering opposing pairs of bus bars **1296** and **1298**, serially, e.g., in a clockwise, counter clockwise, or random manner, rather than energizing all the bus bars at simultaneously. In another example, referring to FIG. **10**, bus bars on opposing corners of the EC device may be powered as pairs, serially, where only two bus bars are energized at any single moment in time. Using this driving algorithm, hot

22

spots can be further avoided, and, because only opposing single pairs of bus bars are energized at any one time, the ends of the bus bars can be closer to each other, even overlapping (i.e., running parallel, side by side but on different electrodes) because only opposing pairs are energized in any single moment in time. Opposing pairs need not be used; the bus bar pairs may include adjacent bus bars, e.g., those described in relation to FIG. **10**, where the ends of the bus bars are spaced appropriately to avoid hot spots.

Methods of Fabrication

FIG. **14** shows a flowchart depicting a method of fabricating an optically switchable device. The method **1400** shown in FIG. **14** describes a method of fabricating a general optically switchable device. Method **1400** may be used to fabricate any of the optically switchable devices including an angled bus bar disclosed herein. The details of forming the optically switchable device would depend on the specific optical device being fabricated, however. For example, method **1400** may be used to fabricate the electrochromic lite **800** shown in FIG. **8A** or the electrochromic lite **850** shown in FIG. **8C** and associated bus bars. Further details regarding the fabrication of electrochromic devices and lites are disclosed in U.S. patent application Ser. No. 12/645,111, filed Dec. 22, 2009, U.S. patent application Ser. No. 12/645,159, filed Dec. 22, 2009, U.S. patent application Ser. No. 12/772,055, filed Apr. 30, 2010, U.S. patent application Ser. No. 12/772,075, filed Apr. 30, 2010, U.S. patent application Ser. No. 12/814,277, filed Jun. 11, 2010, U.S. patent application Ser. No. 12/814,279, filed Jun. 11, 2010, and U.S. patent application Ser. No. 13/166,537, filed Jun. 22, 2011, all of which are incorporated by reference herein for all purposes.

Starting at operation **1405** of method **1400**, an optically switchable device is fabricated on a surface of a substrate in a single integrated vacuum deposition system. During the fabrication process, the substrate is in a substantially vertical orientation in the integrated vacuum deposition system. The optically switchable device includes a perimeter with at least one corner including a first side, a first vertex joining the first side and a second side, and the second side.

At operation **1410**, a first bus bar is formed on the optically switchable device proximate to the corner. The first bus bar includes a first arm and a second arm having a configuration that substantially follows the shape of the first side, the first vertex, and the second side of the corner. The first bus bar is configured to deliver current and/or voltage for driving switching of the optically switchable device.

At operation **1415**, a second bus bar is formed on the optically switchable device. The second bus bar is configured to deliver current and/or voltage for driving switching of the optically switchable device. In some embodiments, the optically switchable device may include a perimeter with a second corner including a third side, a second vertex joining the third side and a fourth side, and the fourth side. The second bus bar may include a third arm and a fourth arm having a configuration that substantially follows the shape of the third side, the second vertex, and the fourth side of the second corner.

Angled Bus Bar Performance

FIGS. **15-17** show graphs including plots showing the performance of planar bus bars as compared to angled bus bars. FIG. **15** shows a graph including plots of the percent transmittance of an electrochromic lite with planar bus bars and an electrochromic lite with angled bus bars. FIG. **16A** shows a graph including plots of the voltage at different positions on a lower conductive electrode of an electrochromic lite. FIG. **17** shows a graph including plots of the current

density in an electrochromic lite with planar bus bars and an electrochromic lite with angled bus bars.

To generate these data, a tempered 26 inch×30 inch glass lite, having a sodium diffusion barrier and fluorinated tin oxide transparent conducting layer, was first coated with an all solid state and inorganic EC device, as described in U.S. patent application Ser. No. 12/772,055, filed Apr. 30, 2010, U.S. patent application Ser. No. 12/772,075, filed Apr. 30, 2010, U.S. patent application Ser. No. 12/814,277, filed Jun. 11, 2010, U.S. patent application Ser. No. 12/814,279, filed Jun. 11, 2010, and U.S. patent application Ser. No. 13/166,537, filed Jun. 22, 2011, all of which are incorporated by reference herein for all purposes. Prior to deposition, a laser isolation scribe (L1), penetrating the fluorinated tin oxide layer, was formed along the entire length of one of the 26" sides of the substrate, approximately $\frac{5}{8}$ " in from the glass edge. After the EC device coating was deposited, two laser scribes (L2) were formed, down to the glass, one each along the 30" sides of the lite, each scribe $\frac{5}{8}$ " from the glass edge. Another laser scribe (L3), parallel to L1, was formed along the other 26" side opposite L1 and $\frac{5}{8}$ " from glass edge. Between the L1 and the nearest parallel glass edge, an ultrasonic soldered bus bar was fabricated approximately the length of the edge to give a conductive path to the top conductive layer with isolation from the bottom conductive layer due to the L1 scribe. Similarly, a soldered bus bar was placed between the L3 and the nearest parallel glass edge to allow for a conductive path to the bottom conductive layer with isolation from the top conductive layer due to the L3 scribe. This yielded a 26"×30" lite with an active area of 24 $\frac{3}{4}$ "×28 $\frac{3}{4}$ " and a planar bus bar configuration (two parallel bus bars running along opposite edges of the lite).

The device was then run through a full coloration and bleaching cycle using a combined technique of cyclic voltammetry-chronocoulometry (CVCC). During cycling, the device was monitored near the center of the active area (plot 1505 in FIG. 15) midway between the L1 and L3 scribes with a light source/detector pair. An edge of the device was monitored (plot 1510 in FIG. 15) with this configuration as well. The center is the area seen to switch slowest on an electrochromic device with the aforementioned construction. This is due to the current drop off across the device due to the sheet resistance of the transparent electrode layers. The instrumentation used for measurement was an Ocean Optics Tungsten Halogen source and ThorLabs amplified silicon detector.

The device was also probed for voltage against the voltage of the bus bar connected to the lower transparent electrode, in a 9 location grid (3×3 matrix) on the active area of the device, as shown in the graphs in FIG. 16A. The 9 points making up the 3×3 matrix were: a point near the top transparent electrode bus bar (i.e., bus bar, 1605, in FIG. 16), a point midway between the bus bars, and a point near the lower transparent electrode bus bar (i.e., bus bar, 1610, in FIG. 16A), each of these three points taken at positions near each of the 30" edges of the lite and midway between the 30" edges of the lite. Each of the graphs shown in FIG. 16A corresponds to one of these points; the graphs are arranged between bus bars 1605 and 1610 in the manner that the device was probed.

The device was driven by an Arbin BT2000 battery tester which cycled the device as well as monitored the auxiliary voltage signals for the optical and voltage measurements. The parameters for the cycling algorithm were:

1. +4 mV/s ramp from 0V to 1V
2. 1V hold for 10 min
3. Ramp from 1V to -1.3V @ -500 mV/s

4. Ramp from -1.3V to -2.5V @ -6 mV/s
5. Hold at -2.5V for 30 min
6. Ramp at +500 mV/s to 0V
7. Ramp at +4 mV/s to 2V
8. 2V hold for 10 min
9. 1V hold for 10 min

The solid lines in the plots are the results of these tests, plotted as voltage versus time. FIG. 16B shows an enlarged view of the graph from the upper left hand corner of FIG. 16A (i.e., graph 1615), for reference.

Following the initial test, an additional L3 scribe was formed $\frac{1}{2}$ " inside one of the L2 scribes, perpendicular to the original L3, configured so that the bus bar electrically connected to the lower transparent conducting layer could be extended along the 30" side, in an angled configuration (approximately 90 degree angle in this example, and having a "L-shape"). This bus bar was extended with solder that extended approximately halfway across the 30" edge of the lite (extension 1625 of bus bar 1605).

The bus bar electrically connected to the top transparent conducting layer was not extended as described above, but rather replaced by a similar L-shaped bus bar that extended down the L1 side of the lite, just inside the old bus bar, and halfway across the 30" side of the lite, opposite the leg of the first L-shaped bus bar and adjacent to the L2 scribe (extension 1630 of bus bar 1610). This bus bar was constructed of copper tape, as ultrasonic solder here would result in an undesirable shunt in the active area of the device. This is because this bus bar resides inside the first bus bar, on the device side of the L1 isolation scribe, and therefore must not penetrate the device stack.

The device was monitored in the same manner at the same locations described in the setup above. That is, the device was run through a full coloration and bleaching cycle using a combined technique of CVCC. During cycling, the device was monitored near the center of the active area (plot 1515 in FIG. 15) midway between the L1 and L3 scribes with a light source/detector pair. An edge of the device was monitored (plot 1520 in FIG. 15), with this configuration as well. The device was also probed for voltage against the voltage of the bus bar connected to the lower transparent electrode, in a 9 location grid (3×3 matrix) on the active area of the device, as shown in the graphs in FIG. 16A. The dotted lines in the graphs are the results of these tests, plotted as voltage versus time. Again, FIG. 16B shows an enlarged view of the graph from the upper left hand corner (i.e., graph 1615), for reference.

The L-shaped bus bar configuration significantly improved coloration and bleaching time across the device. One common metric for characterizing switching speed of an electrochromic device is $t_{80\%}$, or time for the device to reach 80% of its coloration in optical density (OD). For the planar bus bar arrangement, the center and edge $t_{80\%}$ were 38.5 minutes and 22.2 minutes, respectively. With the L-shaped bus bars, the times decreased to 21.5 minutes and 11.5 minutes, respectively, with the larger percent increase occurring at the edge. Moreover, the device range increased and became more uniform with the improved bus bar configuration. Center OD range increased from 1.07 to 1.26 and edge OD range went from 1.08 to 1.35. The improved coloration can be seen in the I-V curve of the CVCC test (FIG. 15) which is a good proxy for measuring switching performance. As shown in FIG. 17, the device's leakage current changed minimally, while the peak current increased by a factor of about 1.4; plot 1705 is the current density with the planar bus bars, and plot 1710 is the current density with the L-shaped bus bars. The greater increase in peak current

25

density signifies an improvement in ion movement and hence faster coloration. The voltage profile (FIG. 16A) as measured across the device showed improvement in all locations. Voltages switched the device not only faster, but with a larger, more uniform range.

Although the foregoing embodiments have been described in some detail to facilitate understanding, the described embodiments are to be considered illustrative and not limiting. It will be apparent to one of ordinary skill in the art that certain changes and modifications can be practiced within the scope of the appended claims.

What is claimed is:

1. An electrochromic device comprising:

a rectangular transparent substrate;

a first transparent conductive layer disposed on the rectangular transparent substrate, the first transparent conductive layer comprising at least one first L-shaped bus bar; and

a second transparent conductive layer disposed on the rectangular transparent substrate, the second transparent conductive layer comprising at least one second L-shaped bus bar;

wherein each of the at least one first L-shaped bus bar and the at least one second L-shaped bus bar comprises only two arms, the two arms having different lengths, and

wherein the first transparent conductive layer is disposed on the second transparent conductive layer, such that a perimeter of the first transparent conductive layer is inside a perimeter of the second transparent conductive layer.

2. The electrochromic device of claim 1, wherein a length of each arm of the at least one first L-shaped bus bar and the at least one second L-shaped bus bar is between about 10% and about 65% of a length of a respective side of the rectangular transparent substrate on which the arm is disposed.

3. The electrochromic device of claim 2, wherein the at least one second L-shaped bus bar is disposed on the second transparent conductive layer, between the perimeter of the first transparent conductive layer and the perimeter of the second transparent conductive layer.

4. The electrochromic device of claim 1, wherein a length of each arm of the at least one first L-shaped bus bar and the at least one second L-shaped bus bar is between about 10% and about 40% of a length of a respective side of the rectangular transparent substrate to which the arm is disposed.

5. The electrochromic device of claim 1, wherein the at least one first L-shaped bus bar is configured to operate at a first polarity and the second at least one L-shaped bus bar is configured to operate at a second polarity opposite the first polarity.

6. The electrochromic device of claim 1, wherein the electrochromic device is disposed in a window of a building.

7. The electrochromic device of claim 1, wherein the at least one first L-shaped bus bar and the at least one second L-shaped bus bar are in (A) a symmetrical relationship to each other or (B) an asymmetrical relationship to each other.

26

8. The electrochromic device of claim 1, wherein the electrochromic device is solid state and inorganic.

9. An electrochromic device comprising:

a rectangular transparent substrate;

a first transparent conductive layer disposed on the rectangular transparent substrate, the first transparent conductive layer comprising a first pair of L-shaped bus bars; and

a second transparent conductive layer disposed on the rectangular transparent substrate, the second transparent conductive layer comprising a second pair of L-shaped bus bars, wherein each L-shaped bus bar of the first and second pairs of L-shaped bus bars comprises only two arms,

wherein the first transparent conductive layer is disposed on the second transparent conductive layer, such that a perimeter of the first transparent conductive layer is inside a perimeter of the second transparent conductive layer.

10. The electrochromic device of claim 9, wherein the first pair of L-shaped bus bars are diagonally opposed and the second pair of L-shaped bus bars are diagonally opposed.

11. The electrochromic device of claim 9, wherein the rectangular transparent substrate has one L-shaped bus bar at each of four corners of the rectangular transparent substrate.

12. The electrochromic device of claim 9, wherein:

the first pair of L-shaped bus bars are adjacent, sharing a common side of the first transparent conductive layer with a first gap between ends of arms at the common side of the first transparent conductive layer; and

the second pair of L-shaped bus bars are adjacent, sharing a common side of the second transparent conductive layer with a second gap between ends of arms at the common side of the second transparent conductive layer.

13. The electrochromic device of claim 12, further comprising an offset between the first gap and the second gap and wherein ends of the L-shaped bus bars of the first pair of L-shaped bus bars and the second pair of L-shaped bus bars do not overlap with one another.

14. The electrochromic device of claim 9, wherein the first pair of L-shaped bus bars are configured to operate at a first polarity and the second pair of L-shaped bus bars are configured to operate at a second polarity opposite the first polarity.

15. The electrochromic device of claim 9, wherein the L-shaped bus bars of the first pair of L-shaped bus bars and the L-shaped bus bars of the second pair of L-shaped bus bars are in (A) a symmetrical relationship to each other or (B) an asymmetrical relationship to each other.

16. The electrochromic device of claim 9, wherein the electrochromic device is solid state and inorganic.

17. The electrochromic device of claim 9, wherein the first transparent conductive layer and the second transparent conductive layer are monolithic transparent conductive material layer.

* * * * *